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Fossil Fuel and Carbon Emission

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Learning Outcomes of Lecture

LO1	Evaluate different type of energy sources and understand their trends, which may include conventional and non-conventional options depending on the context and application.

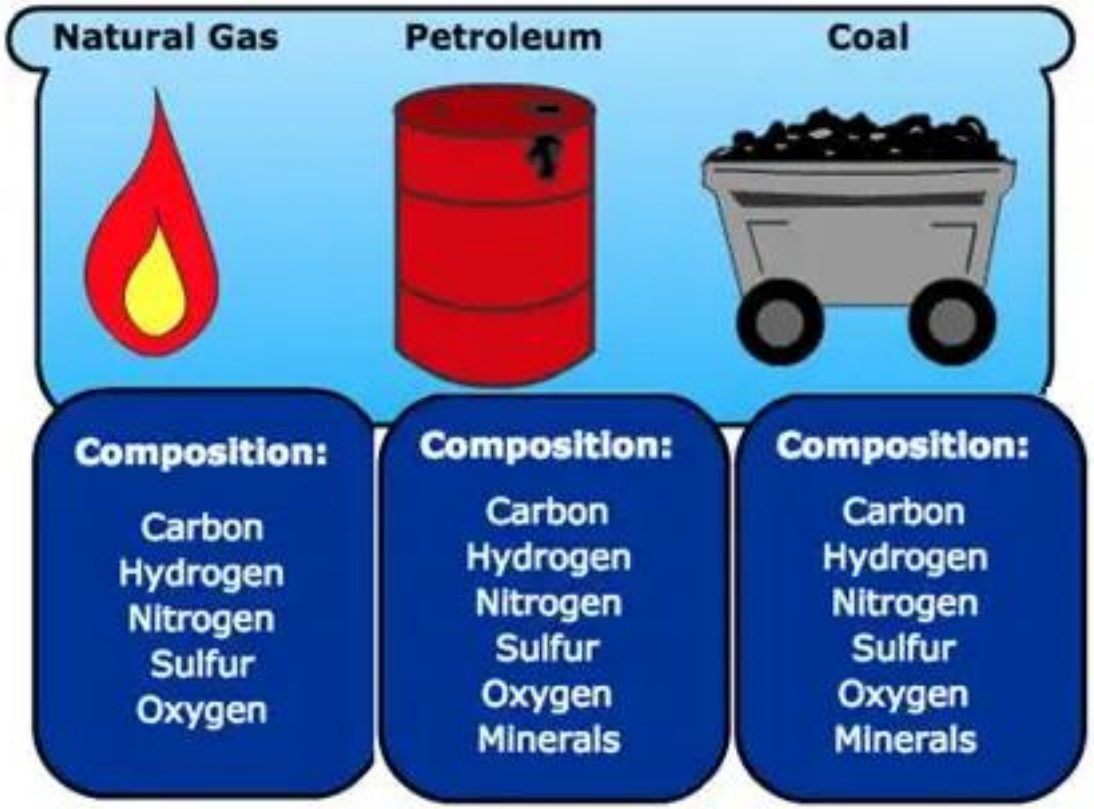
- ❑ Understand the origin, types, and global consumption trends of fossil fuels and their role in meeting global energy demand.
- ❑ Explain the fundamental principles of combustion in solid, liquid, and gaseous fuels, including oxidation reactions and heat release.
- ❑ Identify key atmospheric pollutants and greenhouse gases generated from energy use and combustion processes.
- ❑ Analyse the link between energy consumption, carbon emissions, and climate change, including historical and current emission patterns.
- ❑ Recognise the environmental and societal impacts of fossil fuel use, such as air pollution, acid rain, ozone depletion, and global warming.
- ❑ Evaluate strategies to reduce emissions and environmental impacts, including energy efficiency, cleaner combustion, carbon capture and storage, and alternative energy carriers such as hydrogen.
- ❑ Appreciate the role of international and national climate policies in driving the transition towards low-carbon and sustainable energy systems.

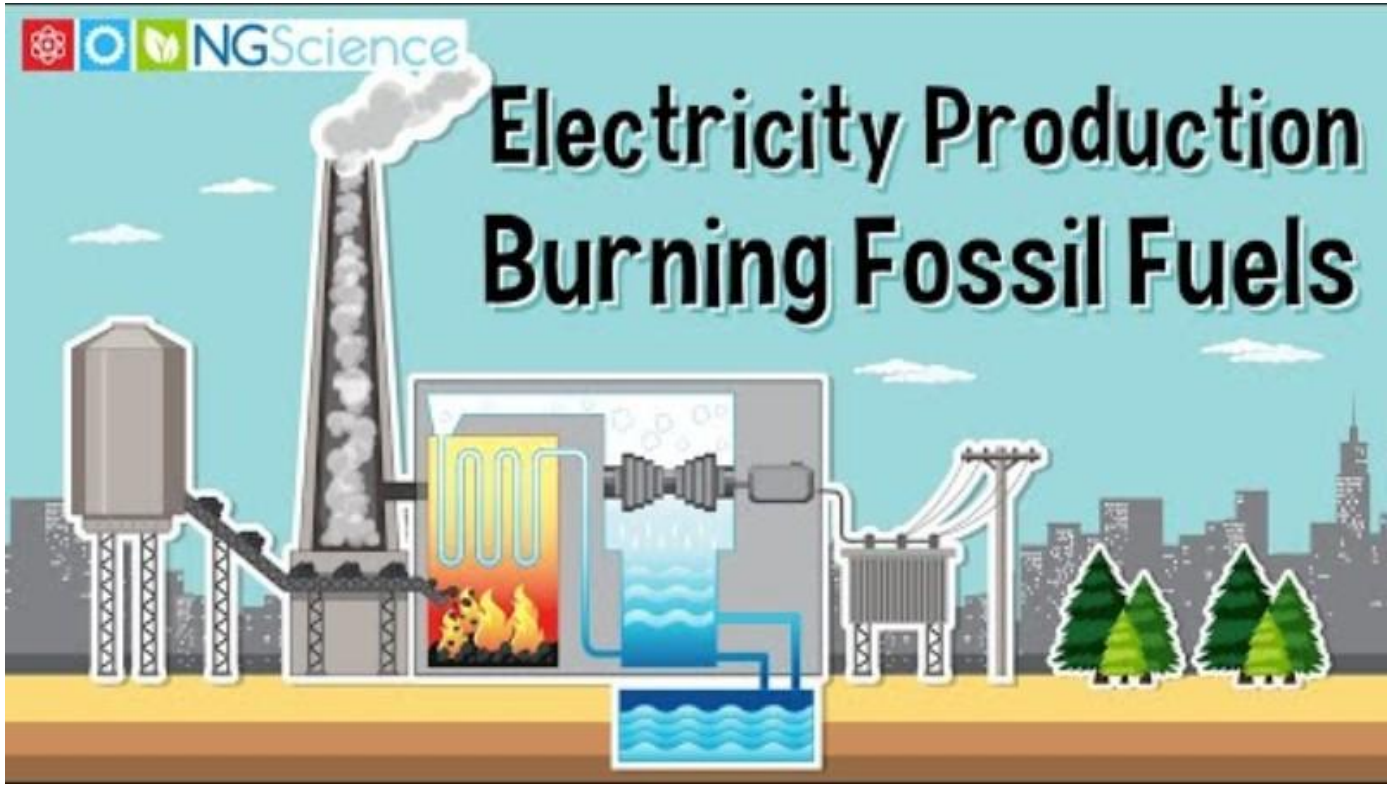
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Fossil fuels

Definition and Formation

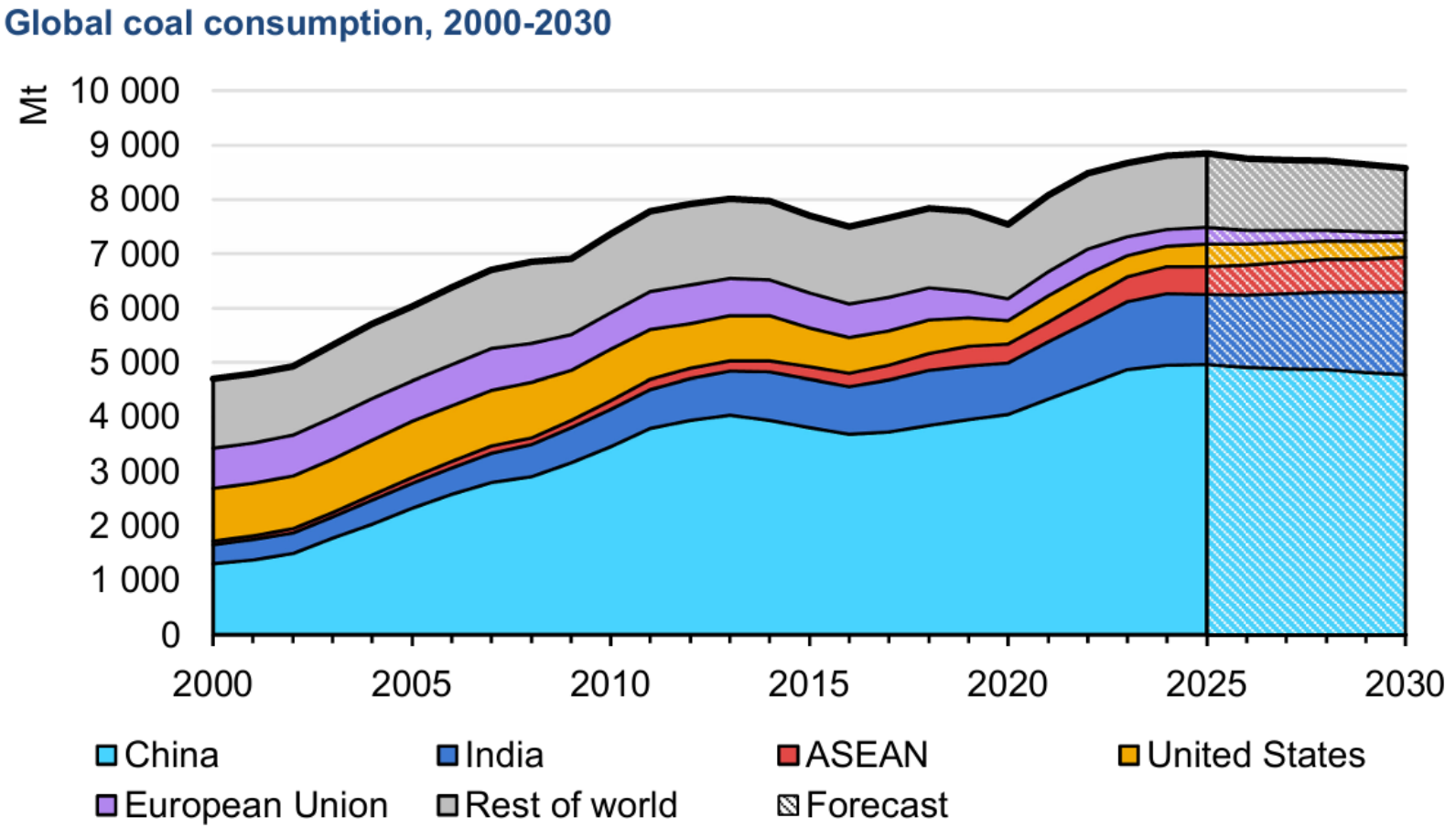
Fossil fuels are **flammable carbon compounds** formed naturally in the Earth's crust from the buried remains of prehistoric organisms, including plants, animals, and microorganisms. This process occurs over millions of years under geological conditions involving heat and pressure. The primary types of fossil fuels include:





Trends of Energy uses by type: Coal

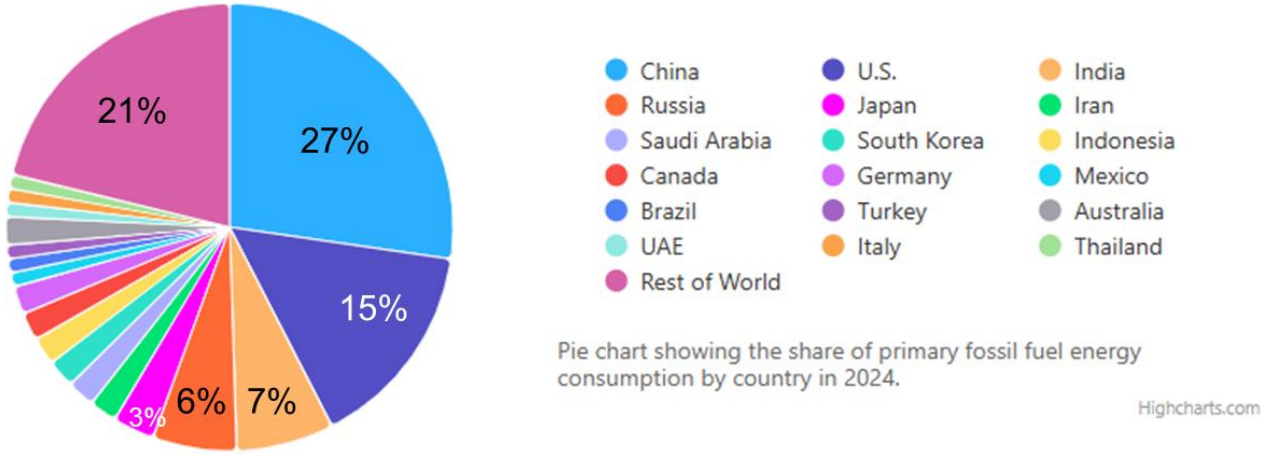
- Global coal demand is expected to effectively plateau over the coming years, showing a very gradual decline through to 2030 in our latest forecast.
- By that year, consumption is forecast to ease by 3% compared with 2025, taking it below its 2023 level.
- Global power generation from coal is forecast to sink below its 2021 level by the end of this decade.

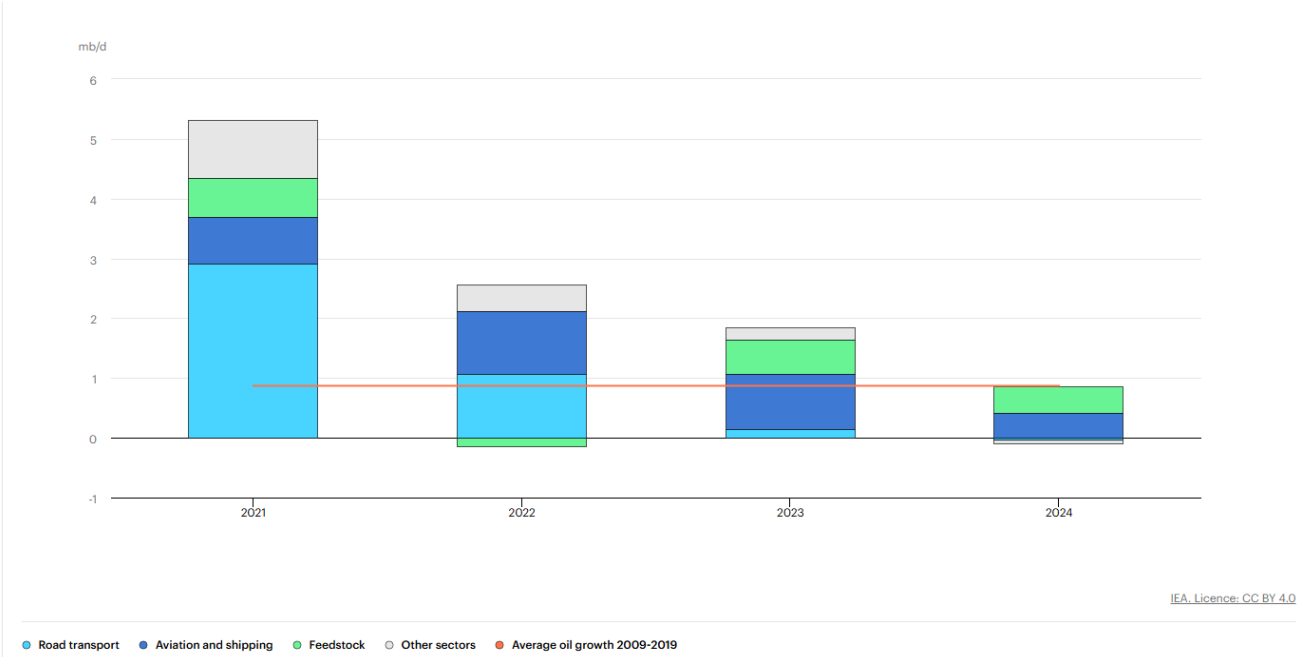


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Trends of Energy uses by type: Oil

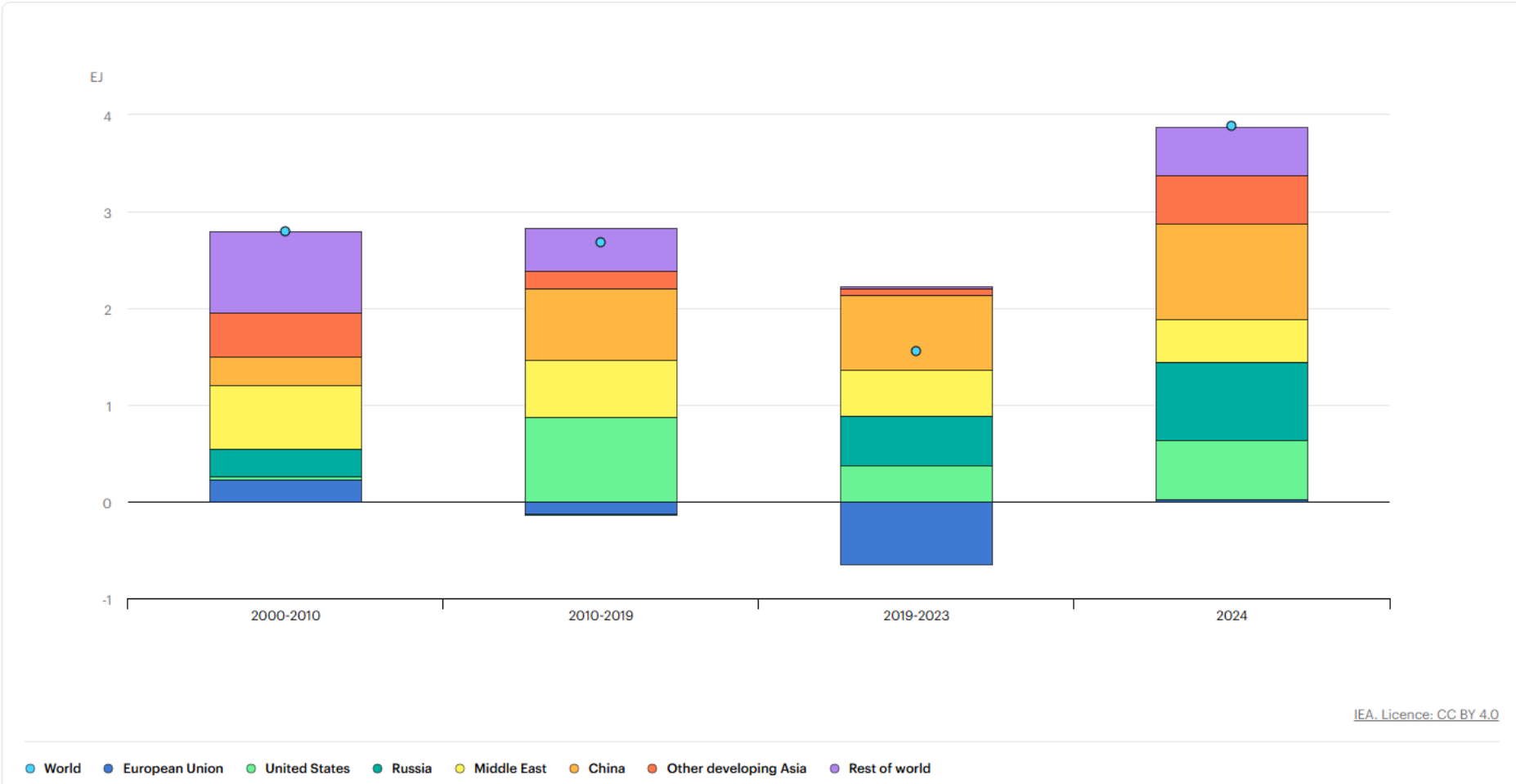
- Global oil demand is forecast to rise by 2.5 mb/d from 2024 to 2030, reaching a plateau around 105.5 mb/d by the end of the decade.
- Annual growth slows from roughly 700 kb/d in 2025 and 2026 to just a trickle over the next several years, with a small decline expected in 2030, based on today's policy settings and market trends.





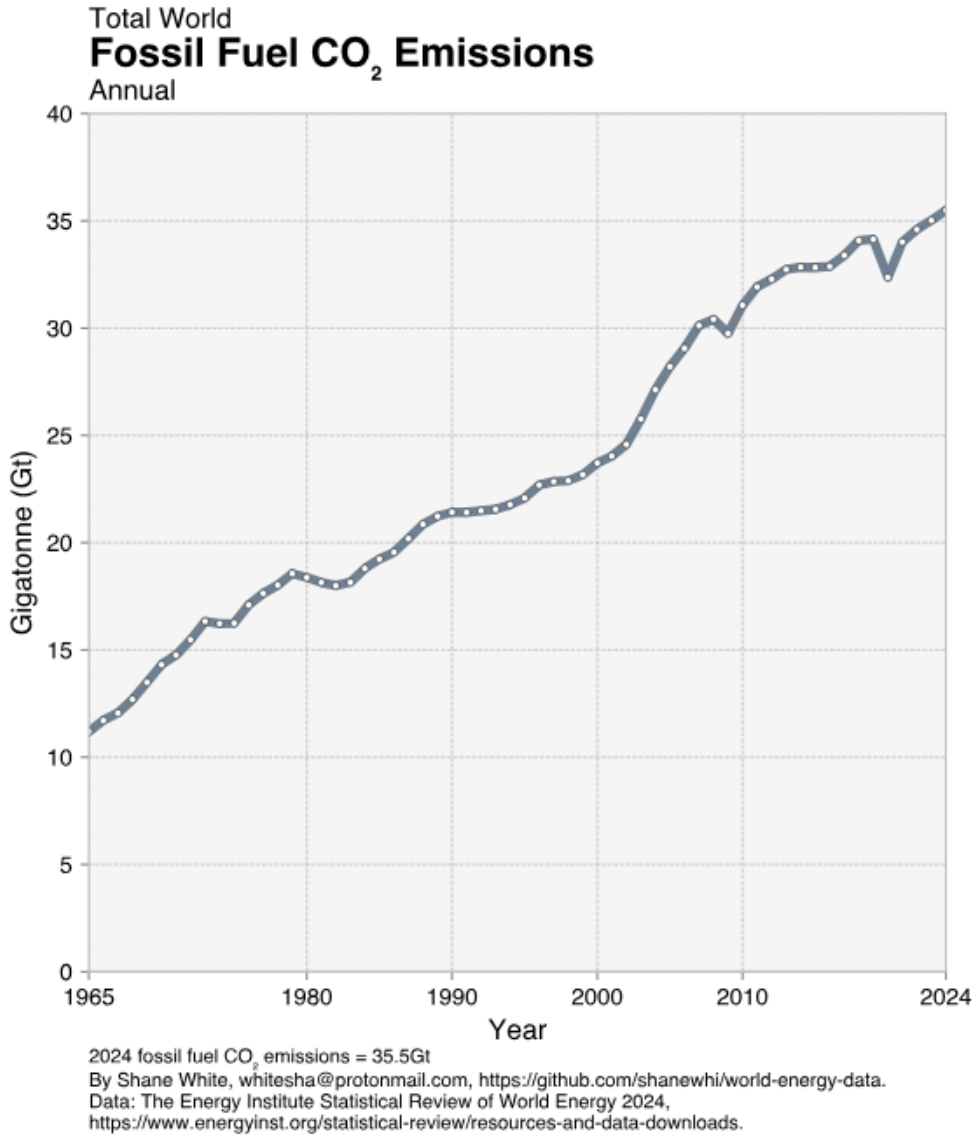
Trends of Energy uses by type: Natural Gas

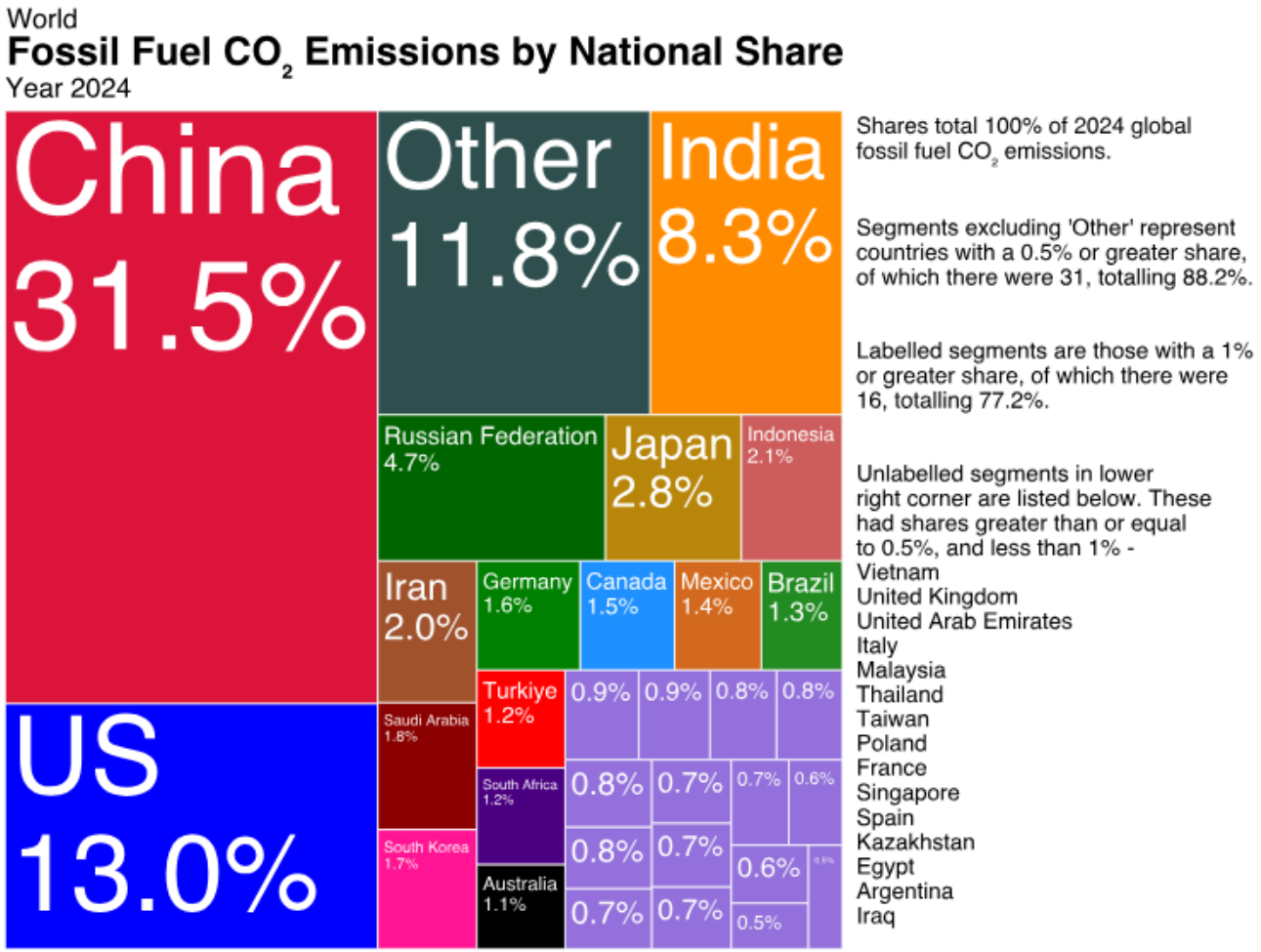
- Gas demand increased by 2.7%, or 115 billion cubic metres (bcm) (equivalent to around 4 EJ) in 2024. This was above the around 2% annual average growth rate from 2010 to 2019 and well above the rate of around 1% between 2019 and 2023, amid the Covid pandemic and global energy crisis.
- Emerging market and developing economies in Asia accounted for around 40% of additional gas demand in 2024 on the back of continued economic expansion.



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Emission from Fossil Fuels by consumers

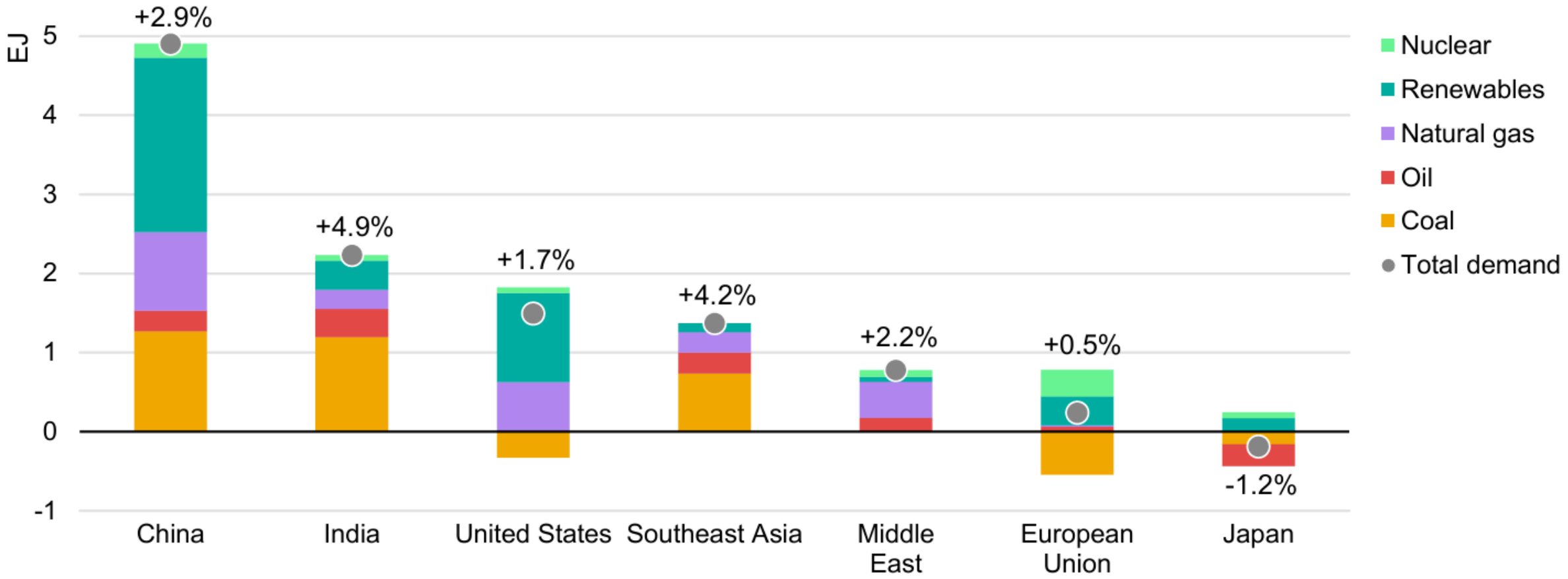




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Change in Global Energy Demand

Change in energy demand, selected regions, 2023-2024



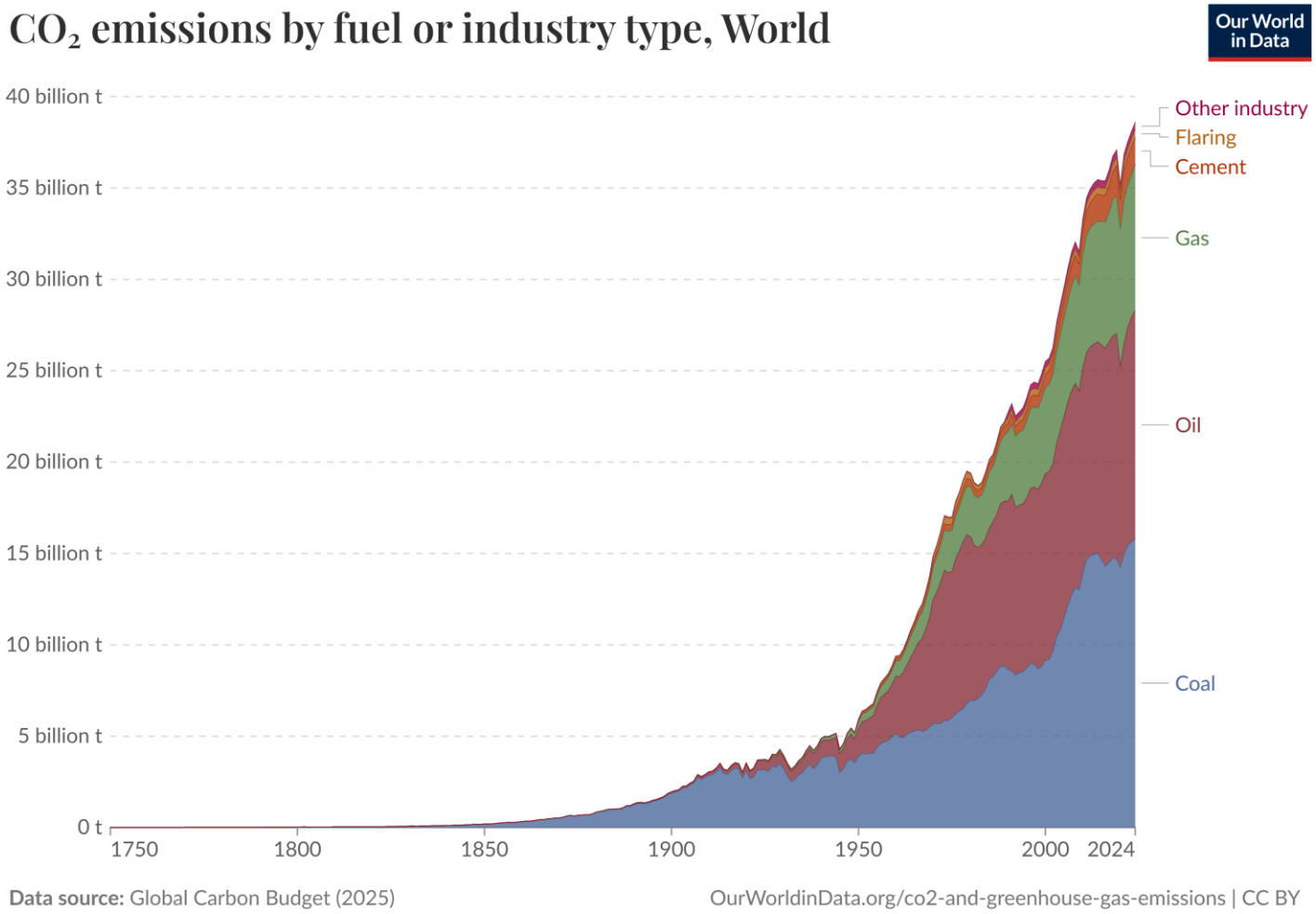
In 2023–2024, energy demand grew in all major regions, driven primarily by **natural gas and renewables**. China, the U.S., and India accounted for the largest absolute increases, though per capita demand remains far higher in developed economies

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Carbon Dioxide Emissions

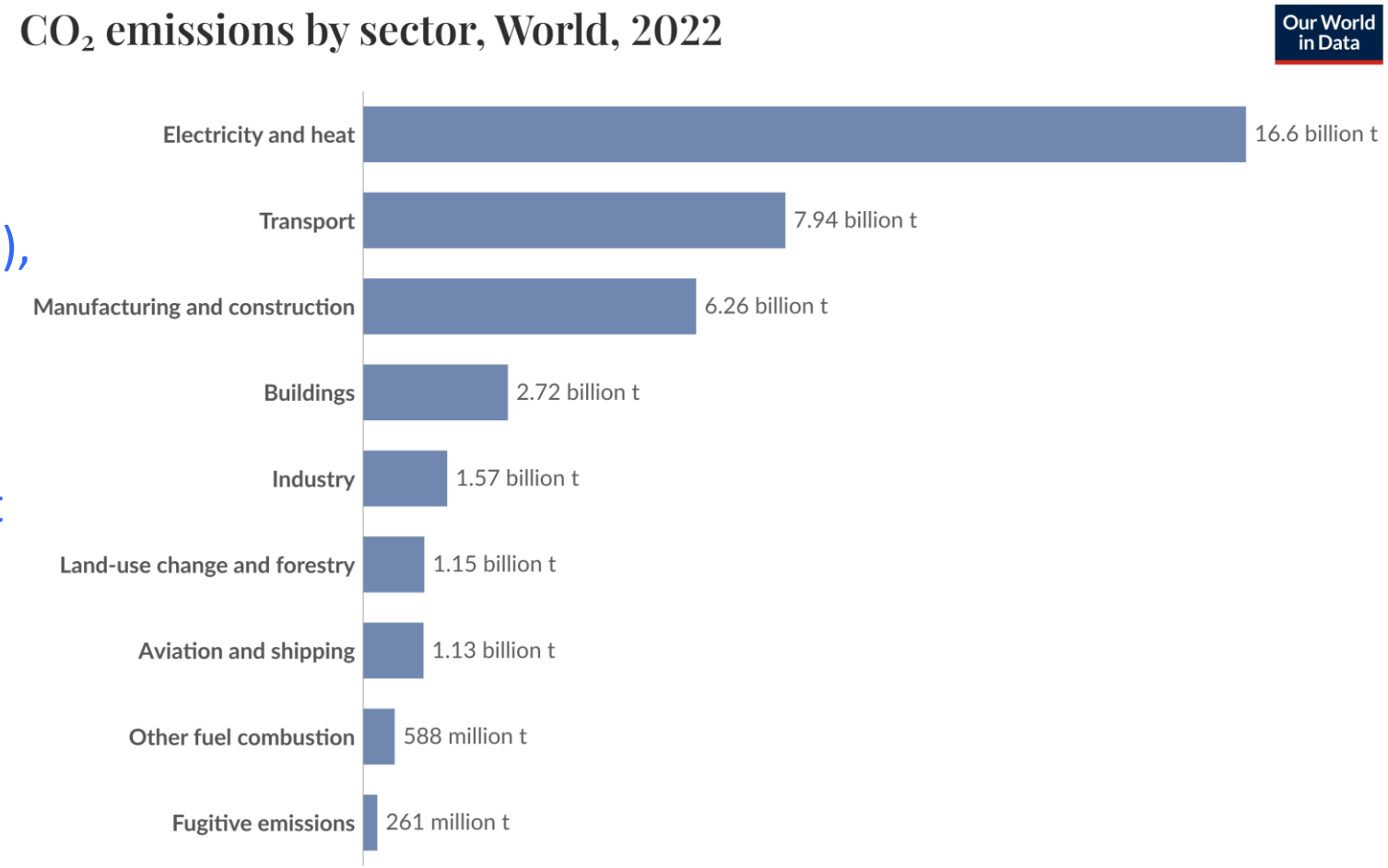
- **Global CO_2 emissions** is showing the overwhelming and persistent dominance of **coal** as the primary driver of CO_2 emissions from fossil fuels throughout modern industrial history.
- Even in the fragmented view provided, the values under "Coal" rise steadily from 0.0 to a range of **0.6 – 0.9 billion tonnes** (and likely far higher in later years), while emissions from Oil and Gas remain at 0.0 in this early timeframe (1750–2017 snippet).
- This highlights that the industrial era's carbon footprint was built on coal, establishing it as the historical foundation of the climate crisis. The recent addition of oil and gas has compounded the problem, but coal's early and sustained role makes its phase-out critically important for mitigation.



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Greenhouse Gas Emissions

- The attached chart shows that electricity and heat generation is by far the largest source of global CO_2 emissions (16.6 billion tonnes in 2022), accounting for nearly double the emissions of the next largest sector, transport (7.94 billion tonnes). Together, these two sectors represent the primary focus for global decarbonization efforts.
- Meanwhile, sectors like buildings, industry, and aviation also contribute significantly, highlighting the need for targeted reductions across multiple areas of the economy.



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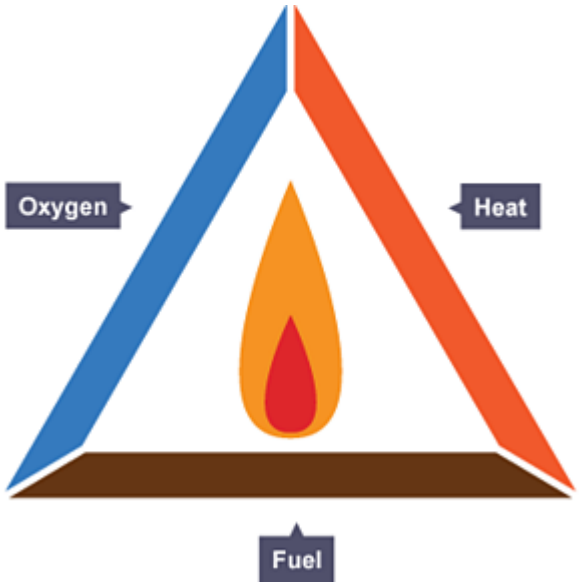
How CO2 is generated? What is Combustion?

Combustion is another name for burning

In combustion, **fuel** is **burned** and reacts with **oxygen** to release **thermal energy (heat)**

The fire triangle summarises the three things needed for combustion - **a fuel, heat and oxygen.** If one of these things is removed from a fire, the fire goes out.

Combustion reactions are used to heat our homes, power most cars, and to generate a lot of our electricity.



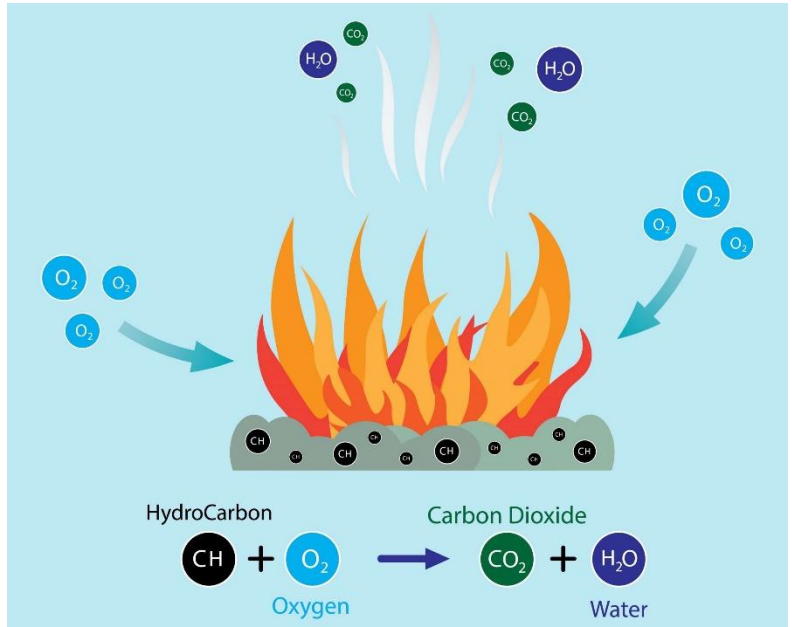


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How CO_2 is generated? What is Combustion?



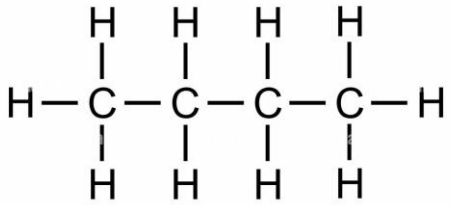
- **Combustion** is an oxidation reaction.
- Many fuels contain carbon and hydrogen atoms. These fuels burn to produce oxides, carbon dioxide and water.
- Water, H_2O, is dihydrogen monoxide.
- Carbon dioxide, CO_2, is a greenhouse gas that causes global warming.



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Combustion Equations: Butane Example

C_4H_{10}.



BUTANE

butane + oxygen → carbon dioxide + water.

$$C_4H_{10} + O_2 \rightarrow CO_2 + H_2O$$

Most combustion reactions produce **carbon dioxide and water**, so these chemicals are written as the products on the **right** of the equation.

Charcoal contains only carbon atoms – what is the combustion equation for charcoal on a BBQ?

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Combustion of Solid Fuels

A combustion process is always exothermic and liberates heat. The reaction taking place is:

Fuel + O_2 -> Products + Heat

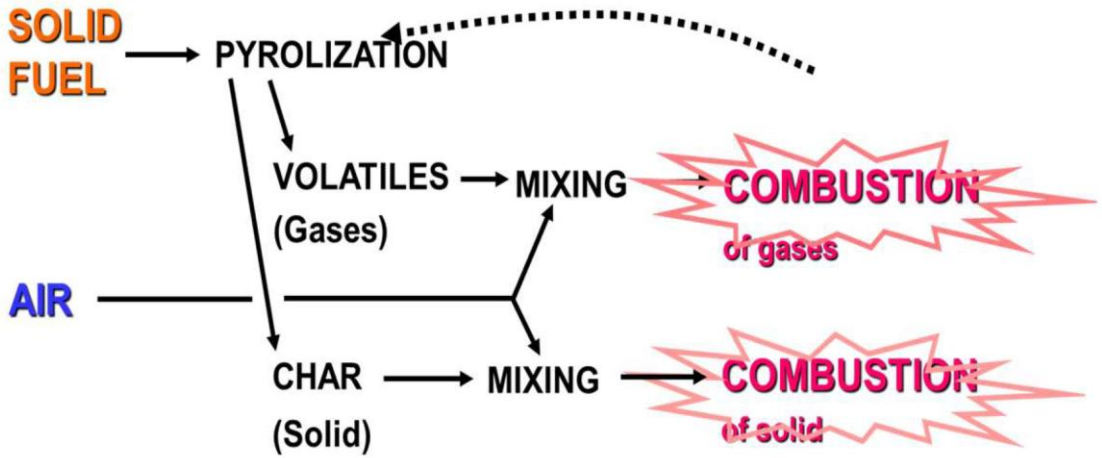
Solid fuels combustion generally goes through four stages:

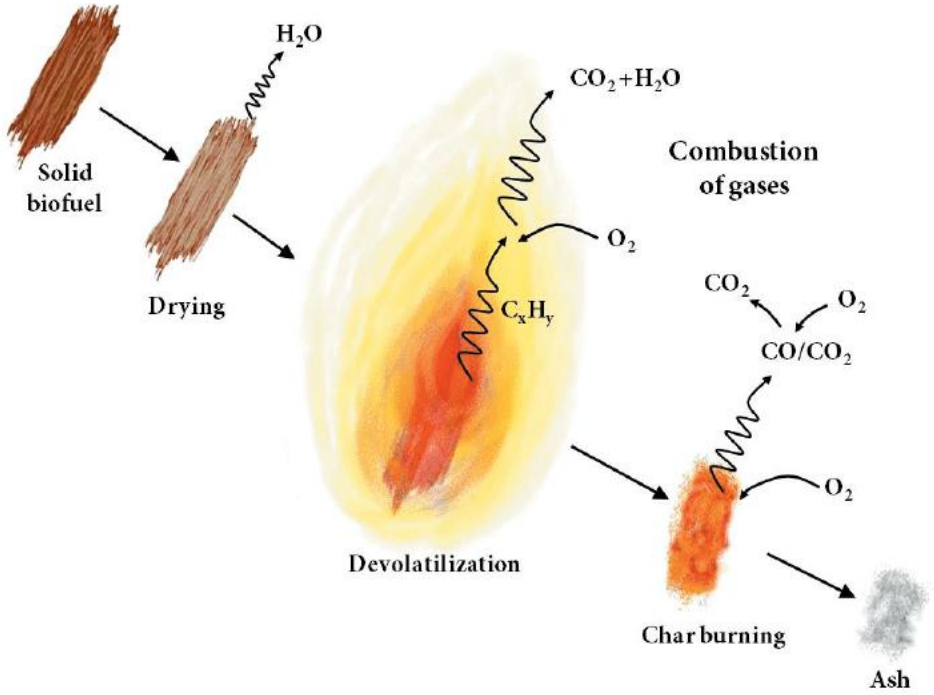
i. Drying: removal of moisture

ii. Devolatilization: decomposition process where volatiles are removed

iii. Burning of volatile substances

iv. Burning of char





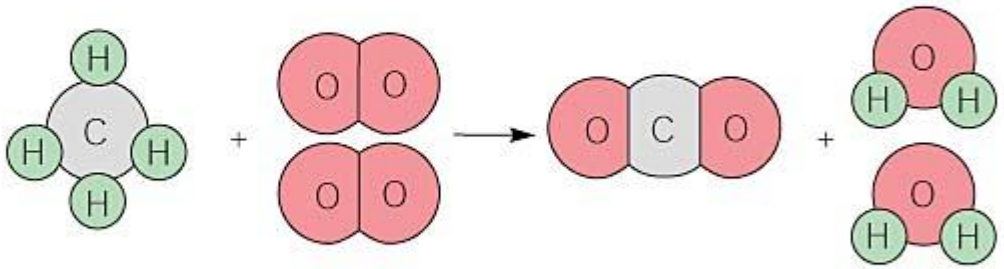
https://www.researchgate.net/publication/260243997_Fate_of_Fuel-Bound_Nitrogen_and_Sulfur_in_Biomass-Fired_Industria

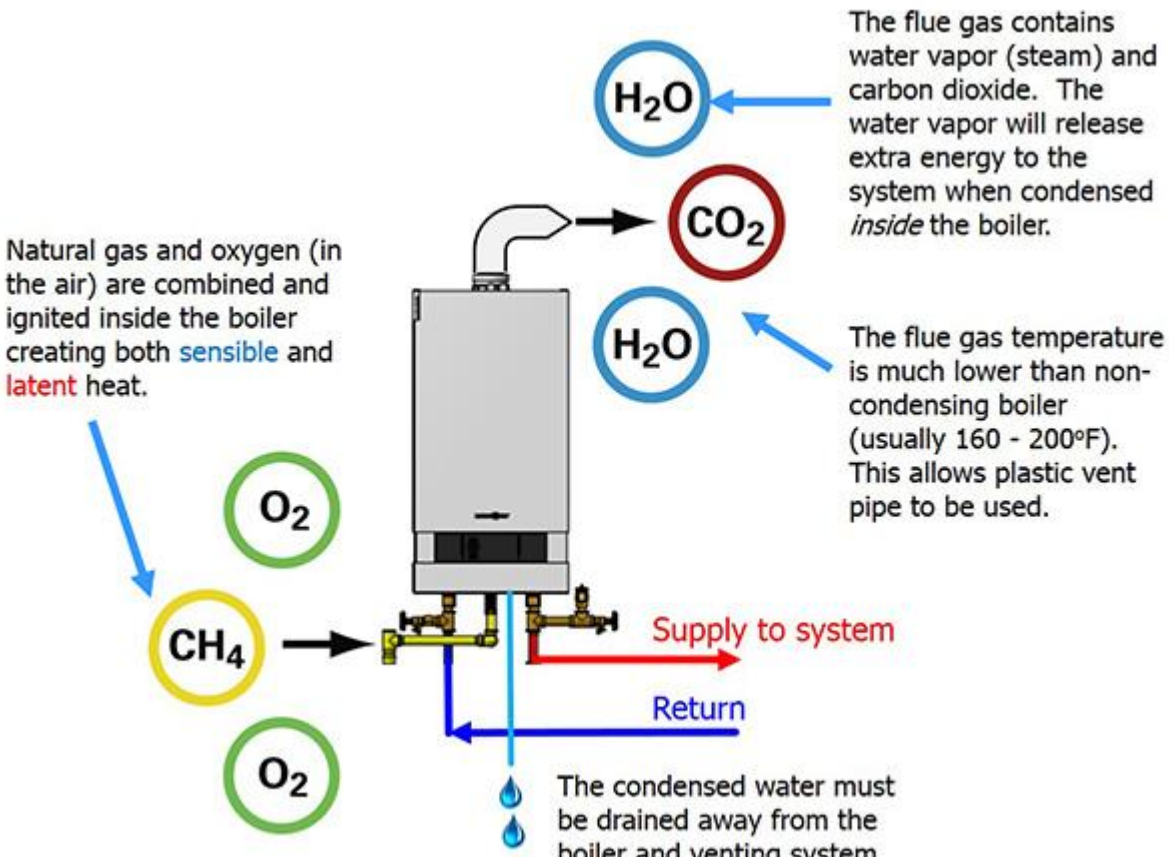
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Combustion of Gaseous Fuels

Gas combustion



Reaction:	Methane reacts with oxygen to yield carbon dioxide and water
Balanced equation:	$CH_4 + 2\ O_2 \longrightarrow CO_2 + 2\ H_2O$
Sketches representing molecules:	
Meaning:	1 molecule of methane + 2 molecules of oxygen → 1 molecule of carbon dioxide + 2 molecules of water



https://www.hpacmag.com/features/dealing-with-the-products-of-combustion-part-ii/

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Combustion of Liquid fuels

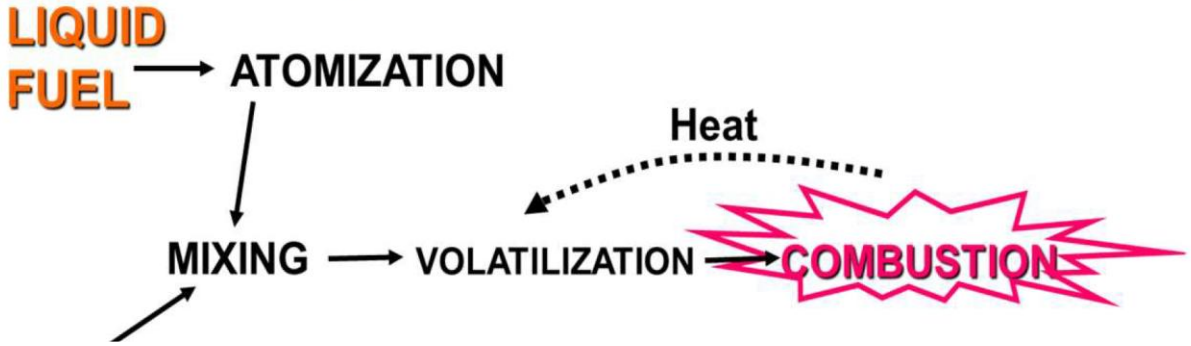
The process of liquid fuel combustion is in 3 stages

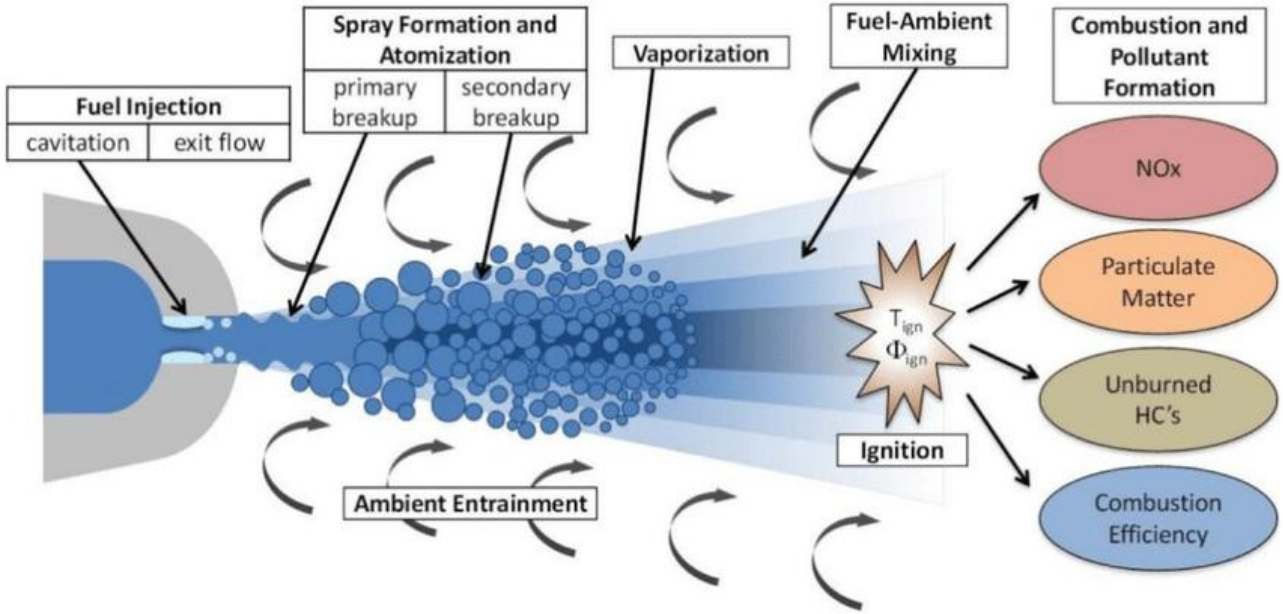
i. First, the liquid fuel is atomized by a spray nozzle to generate fine droplets.

ii. Then, in the high temperature environment of the combustor, those droplets start to vaporize.

iii. Finally, once the temperature is sufficiently high, fuel vapor is auto-ignited and combusted.

Thus, the combustion performance of liquid fuels is largely influenced by the atomization and droplet combustion processes.





https://doi.org/10.1016/j.pecs.2019.03.001

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Gasification

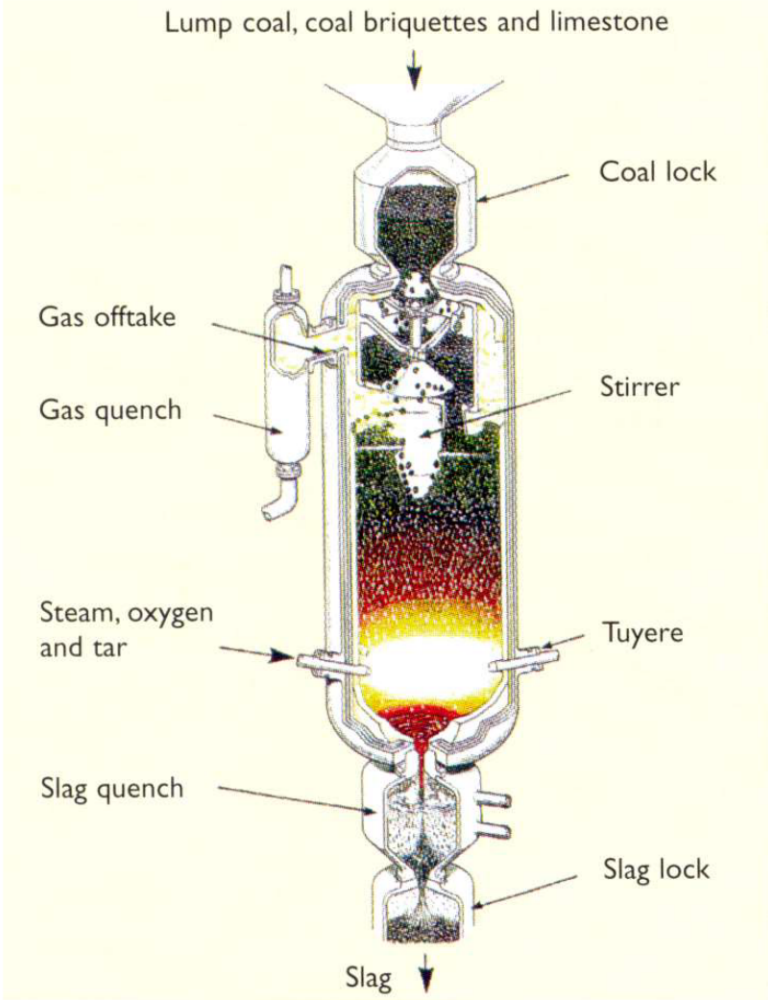
FUEL (*solid*) ⟶ FUEL GAS

Heat, Steam, Oxygen/Air

Representative Fuel Gases from Coal Gasification

	Air/Steam blown	Oxygen/Steam blown
Carbon monoxide	17%	61%
Hydrogen	24%	28%
Nitrogen	41%	-
Carbon dioxide	14%	3%
Methane	4%	8%
Gross Calorific Value	**6.4 MJ/m^3**	**14.2 MJ/m^3**

Natural gas – 38 MJ/m^3



Advantages:

- Fuel gas easy to clean
- Nitrogen in fuel reduced to N_2
- Fuel gas can be burned in gas turbines and internal combustion engines

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Gasification



	Air/Steam blown	Oxygen/Steam blown
Carbon monoxide	17%	61%
Hydrogen	24%	28%
Nitrogen	41%	-
Carbon dioxide	14%	3%
Methane	4%	8%

Convert fuel gas to Hydrogen - for Carbon Capture

$C + H_2O = CO + H_2$ **Gasification Reaction**

$CO + H_2O = CO_2 + H_2$ **Water Gas Shift Reaction**

- Can react with more steam to convert all carbon to CO_2
- CO_2 can then be removed and stored (this is easier than post combustion)
- Hydrogen rich gas can then be combusted giving no CO_2 emissions

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Pyrolysis

FUEL (*solid*) → FUEL GASES, OILS, CHAR

Heat

Advantages:

- Liquid products have potential for transport applications
- Products have high calorific value

Disadvantages

- More than one product – complex process – further refining needed
- Solid char may be undesirable

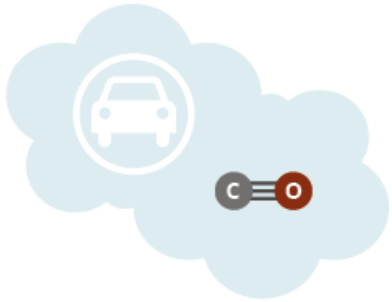


https://www.linkedin.com/pulse/torrefaction-pathway-towards-fungible-biomass-koukoulas-ph-d-/

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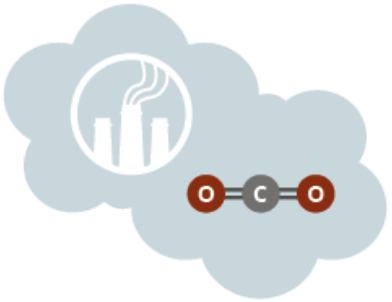
A BRIEF GUIDE TO ATMOSPHERIC POLLUTANTS

A number of different chemical entities, from a range of sources, can contribute towards atmospheric pollution, the consequences of which can include global warming and smog. This graphic looks at a selection of major groups of atmospheric pollutants, their major sources, and their effects.



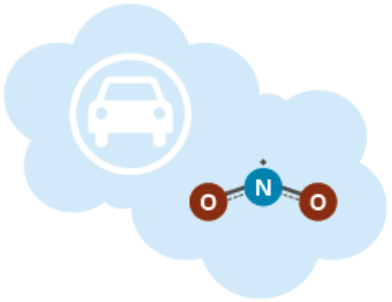
CARBON MONOXIDE

A gas generated by the incomplete combustion of fuels – primarily from road transport. Affects human health, as it reduces oxygen-carrying capacity of the blood. It also reacts with other atmospheric gases to produce ozone.



CARBON DIOXIDE

A gas generated by the burning of fossil fuels in the production of electricity. Also emitted by natural processes. Human emissions are linked with rising atmospheric CO_2 levels and anthropogenic global warming.



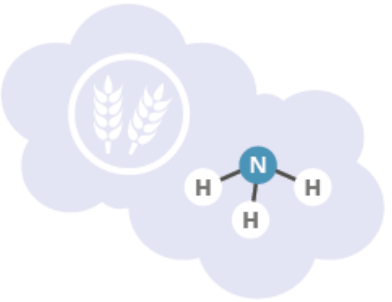
NITROGEN OXIDES

Primarily created by combustion in road transport. Nitrous oxide is an important global warming contributor, whilst nitrogen dioxide is involved in ground-level ozone forming reactions, and is also a component of smog.



SULFUR DIOXIDE

The primary source of sulfur dioxide is the burning of fossil fuels to generate electricity. It can contribute to smog, reacts with water to produce acid rain, and can also cause wheezing and breathing problems for asthmatics.

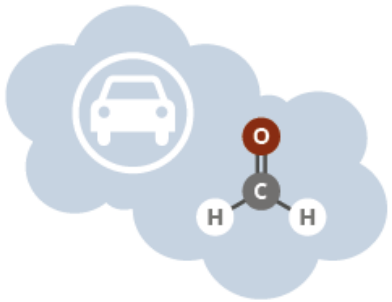


AMMONIA

Ammonia's primary atmospheric source is from its use in agriculture, such as manure & fertilisers. It can react with other pollutants to produce particulate matter. It also has the ability to over-enrich ecosystems with nitrogen.

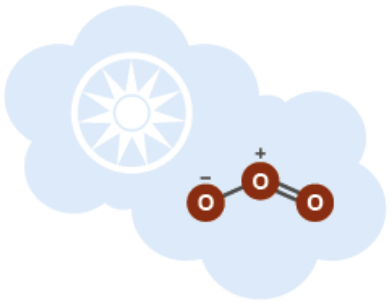
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A BRIEF GUIDE TO ATMOSPHERIC POLLUTANTS



VOCs

VOCs (volatile organic compounds) are emitted naturally by vegetation. Amongst significant human sources is road transport, as well as solvents. They can contribute to formation of ground-level ozone and smog.



OZONE

The ozone layer shields us from UV radiation, but ground-level ozone is a major pollutant. It's formed from other pollutants in the presence of sunlight. Ozone is a major component of smog, and can also cause health effects.



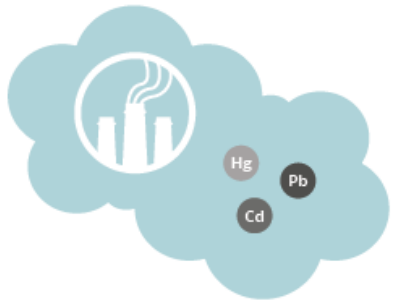
POPs

POPs (persistent organic pollutants) are volatile chemicals released into the atmosphere, often from agricultural or industrial uses. They persist in the environment and can have health effects on both wildlife & humans.



PARTICULATE MATTER

Particulate matter is composed of a huge number of different components. Some are directly emitted, while others are generated by reactions in the atmosphere. They cause haze and can also cause lung problems if inhaled.



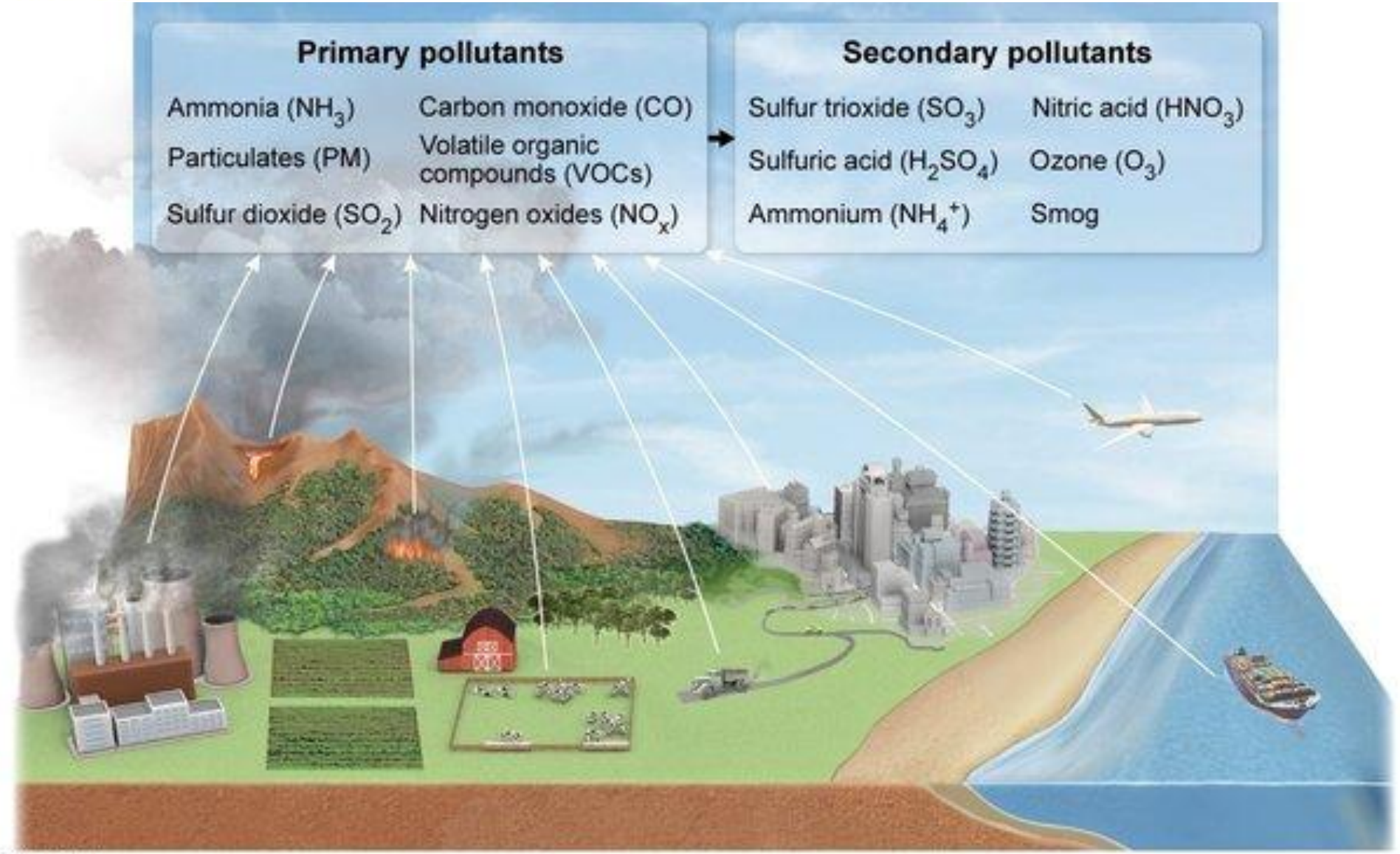
HEAVY METALS

Heavy metals are released into the atmosphere from a range of sources, including burning of fossil fuels and road transport emissions. Some, such as mercury and lead, have toxic health effects in humans.

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Primary and Secondary Pollutants



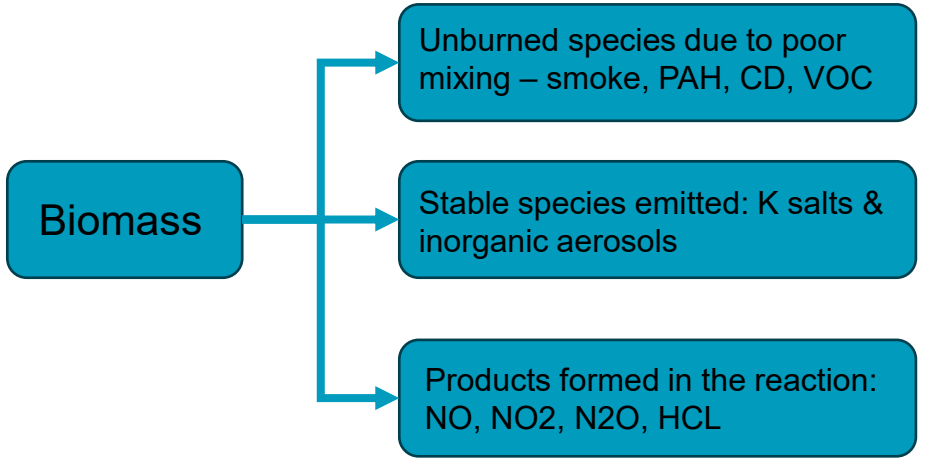
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• **Pollutants from incomplete combustion**: Incomplete combustion produces products such as carbon (C) and carbon monoxide (CO) as well as water and carbon dioxide. The burning flame is typically yellow or orange and there is smoke

• **Pollutants from contaminants in fuel**

• **Formation of oxides of nitrogen**

• **Greenhouse gases:** carbon dioxide, methane, ozone, nitrous oxide, chlorofluorocarbons, and water vapor. Water vapor, which reacts to temperature changes, is referred to as a 'feedback', because it amplifies the effect of forces that initially caused the warming.





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Complete combustion takes place in the presence of a sufficient amount of oxygen while an incomplete combustion reaction takes place when there is an insufficient amount of oxygen supply

Pollutants:

- Carbon monoxide (CO)
- Unburnt fuel: smoke/soot particles (+ organic compounds), gaseous hydrocarbons, carbon particles in mineral ash residue

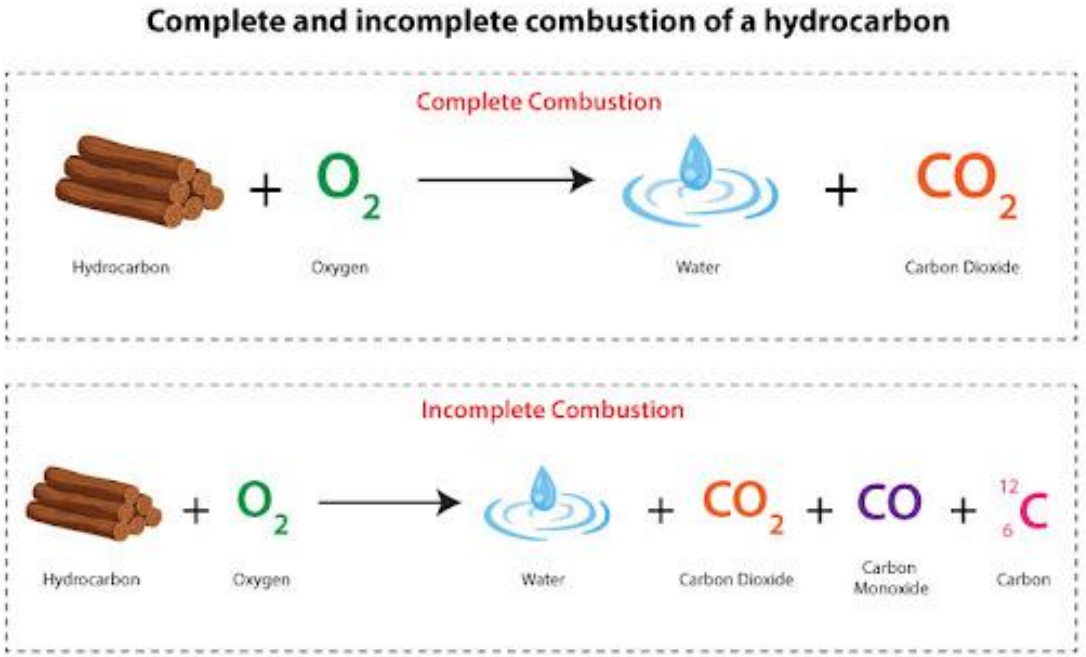
Harmful Effects:

- CO is toxic
- Soot is a nuisance
- Unburnt hydrocarbons (HC) may be toxic or carcinogenic

Control: ensure good combustion conditions:

- sufficient air
- good mixing
- high temperatures
- Remove any remaining materials by catalysts/filters (e.g. in engines)

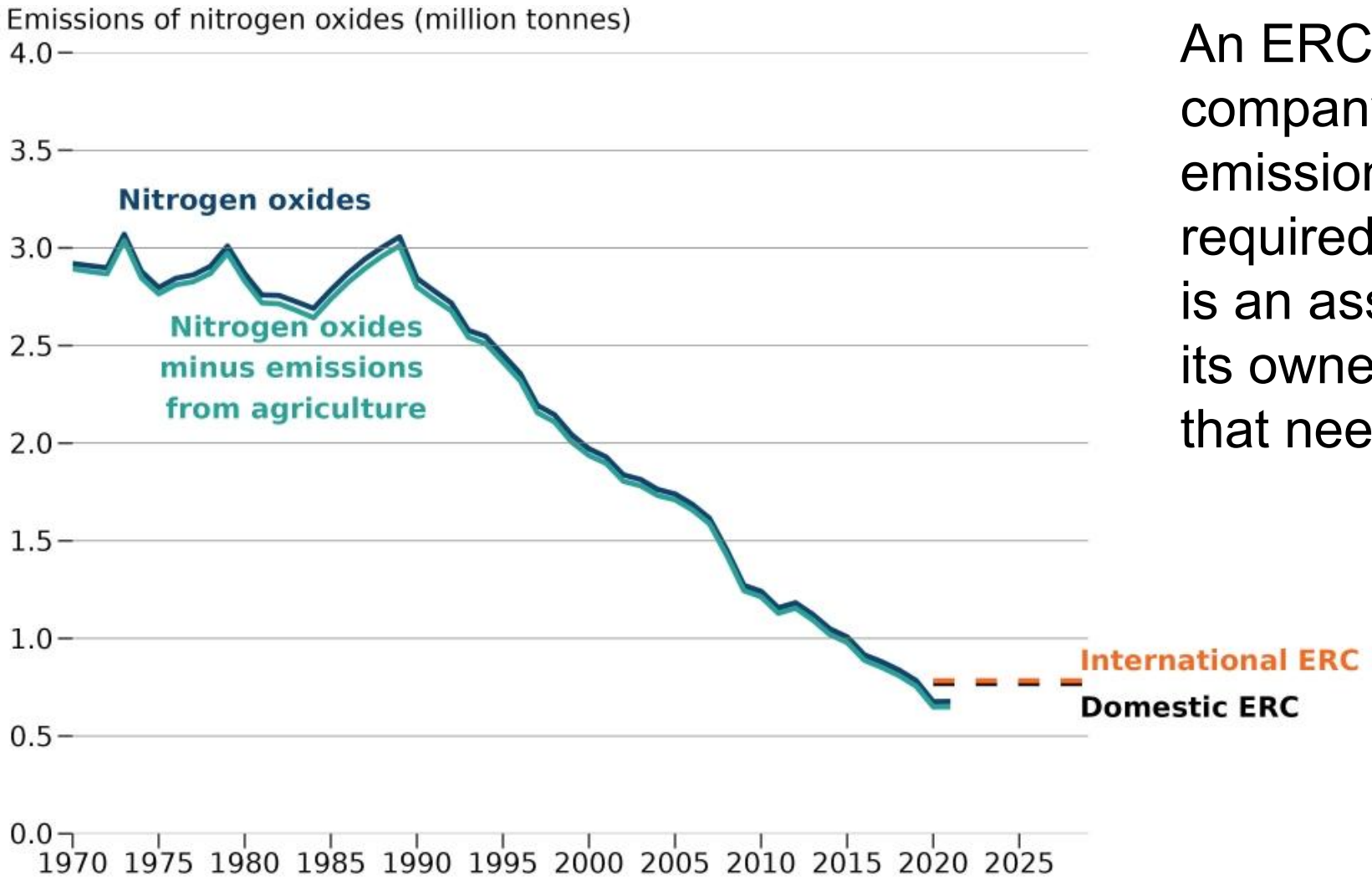
For wastes typical conditions: >6% O_2, >850 °C, > 2 secs





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Emission Reduction Credit (ERC)

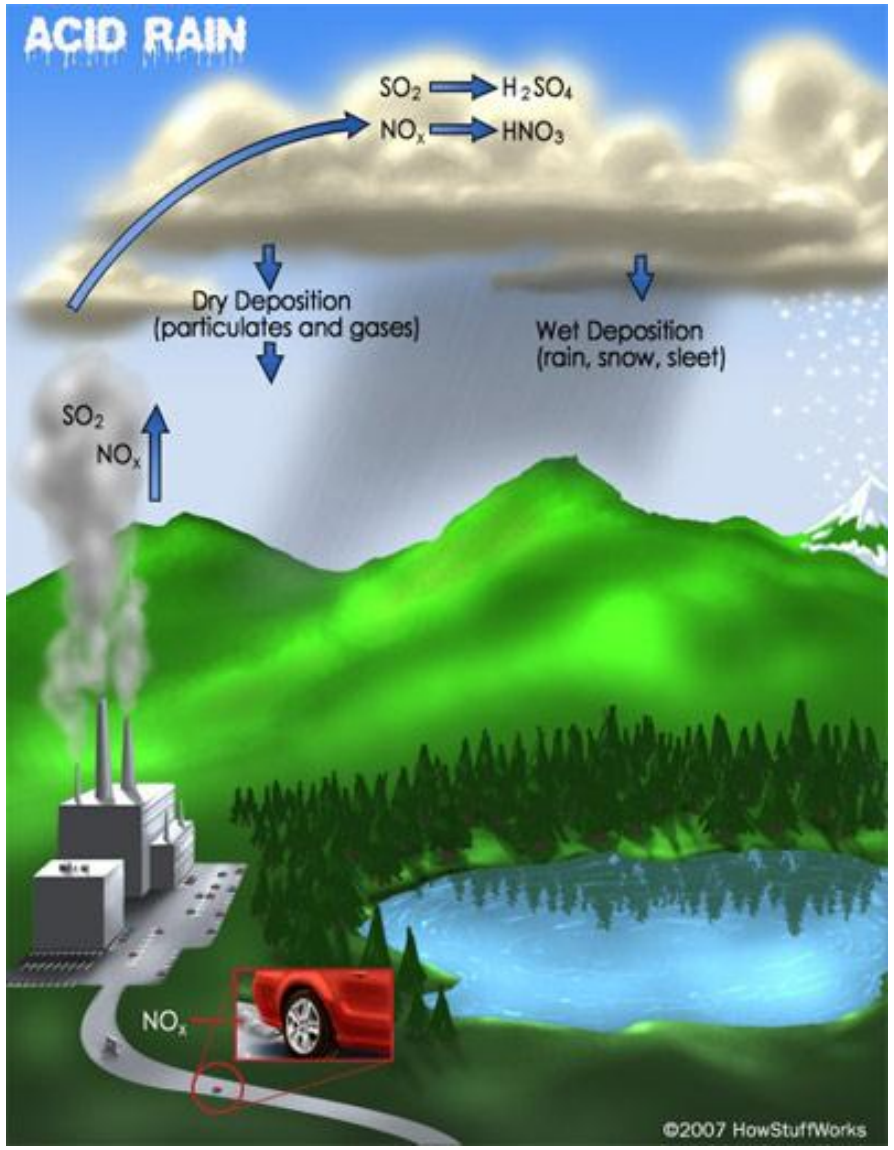


Source: Ricardo Energy & Environment

What is an ERC?
An ERC is a credit earned by a company when it reduces air emissions beyond what is required by permits and rules. It is an asset that can be used by its owner or sold to companies that need emission offsets.

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An Example from History of Climate crises: Acid rain

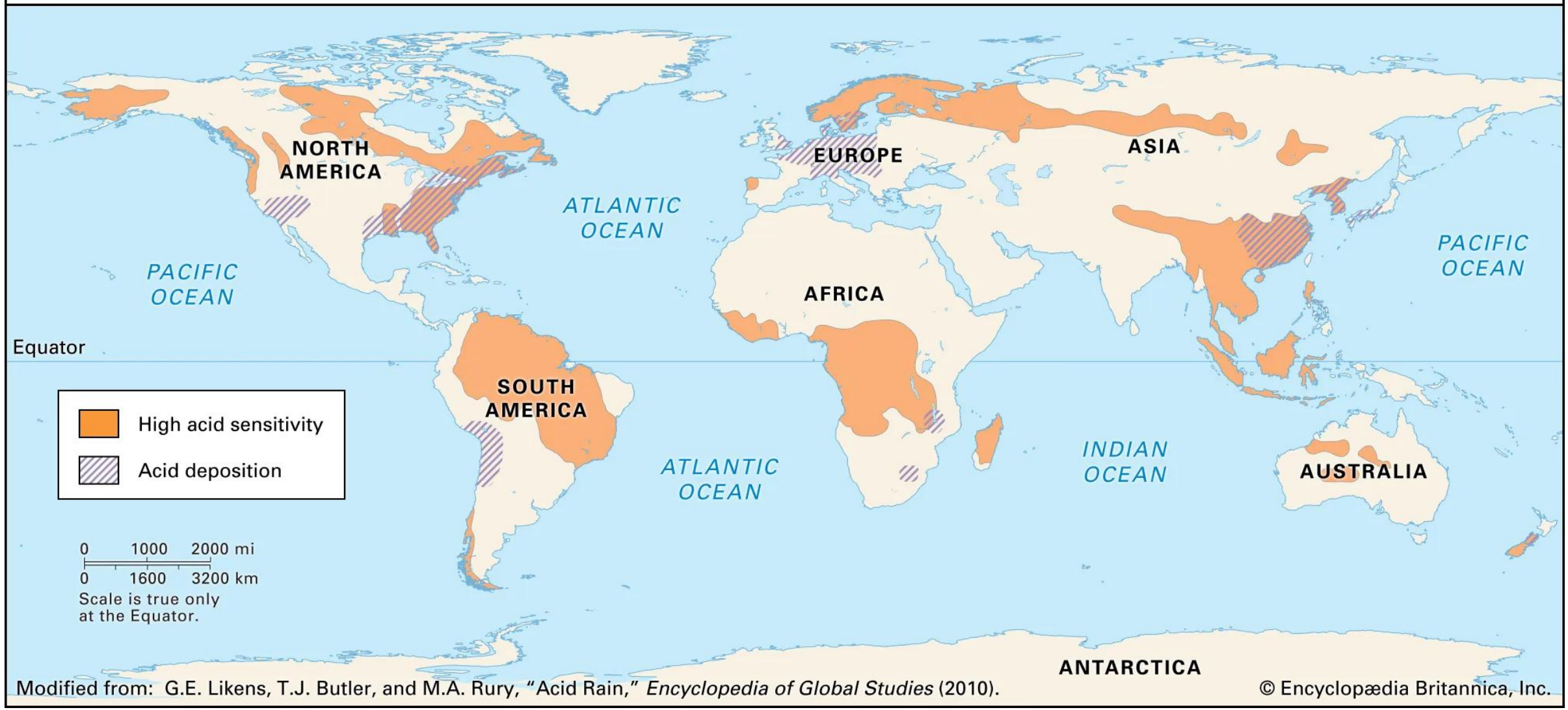




As early as the 1960s, scientists observed evidence of tree damage from acid rain and other environmental pollutants in the Black Forest of Germany. First termed Waldsterben, or tree death, this phenomenon caused damage to nearly half of all trees in the Black Forest by 1990. Acid rain harms wildlife, and while most acid precipitation studies focus on aquatic animals, the forests are not immune to the effects of acid rain.

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ACID SENSITIVE REGIONS

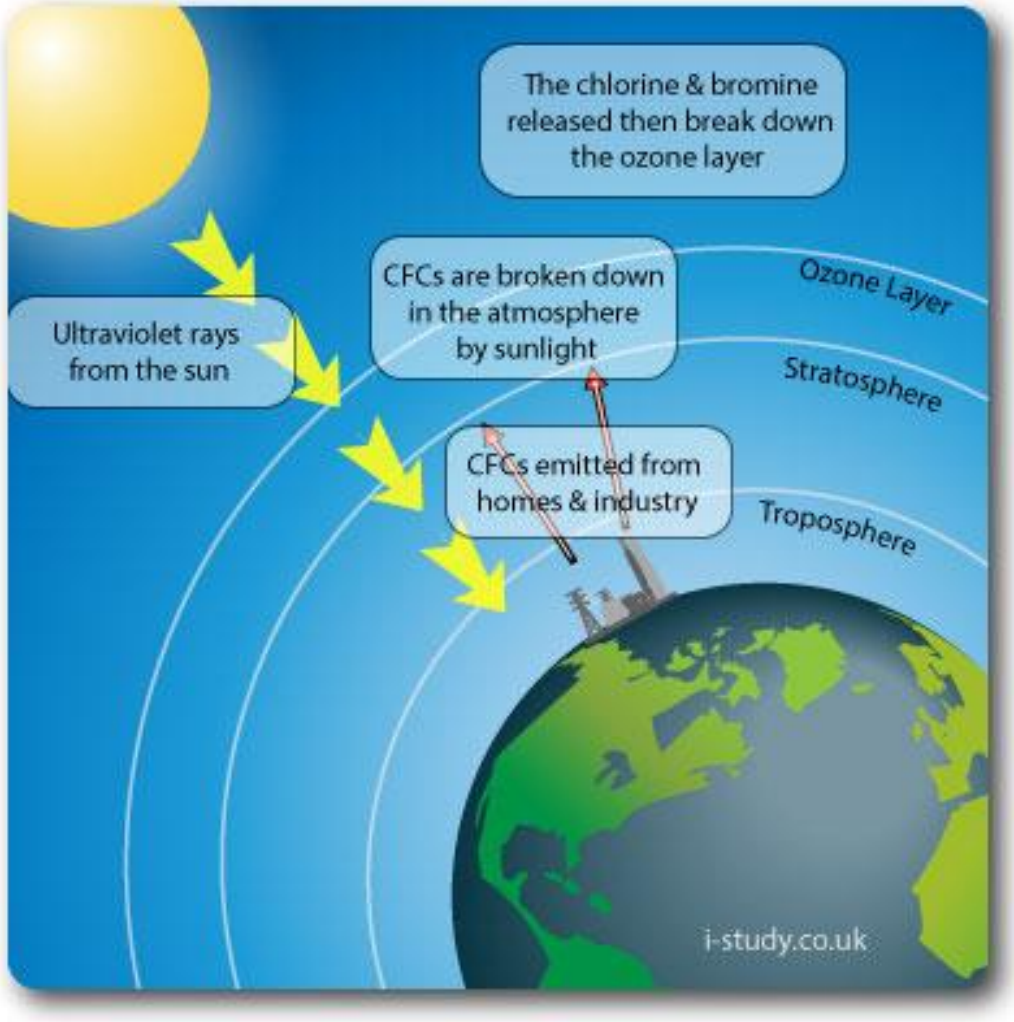


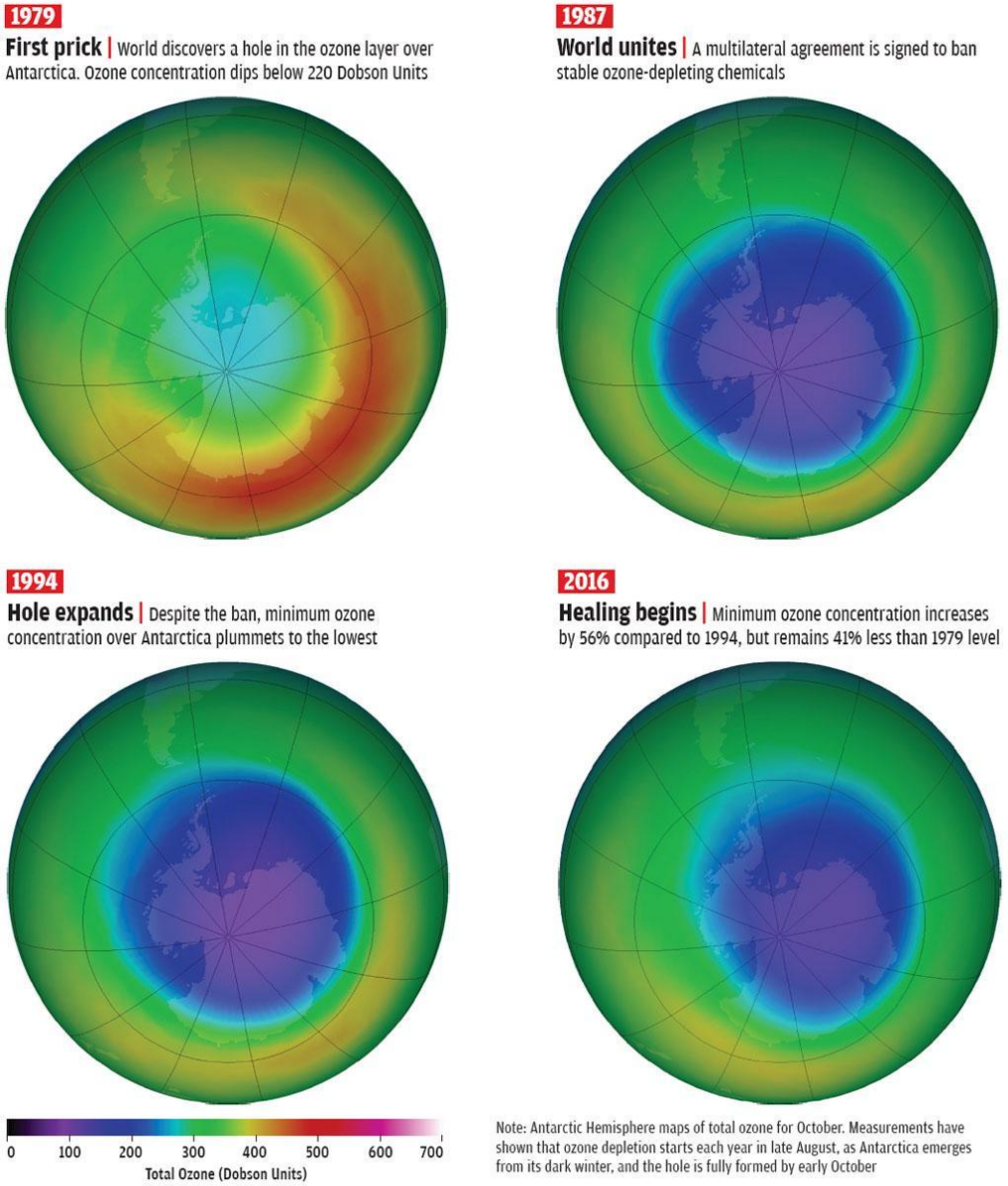
Modified from: G.E. Likens, T.J. Butler, and M.A. Rury, "Acid Rain," *Encyclopedia of Global Studies* (2010).

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Ozone layer Hole





https://ozonewatch.gsfc.nasa.gov/

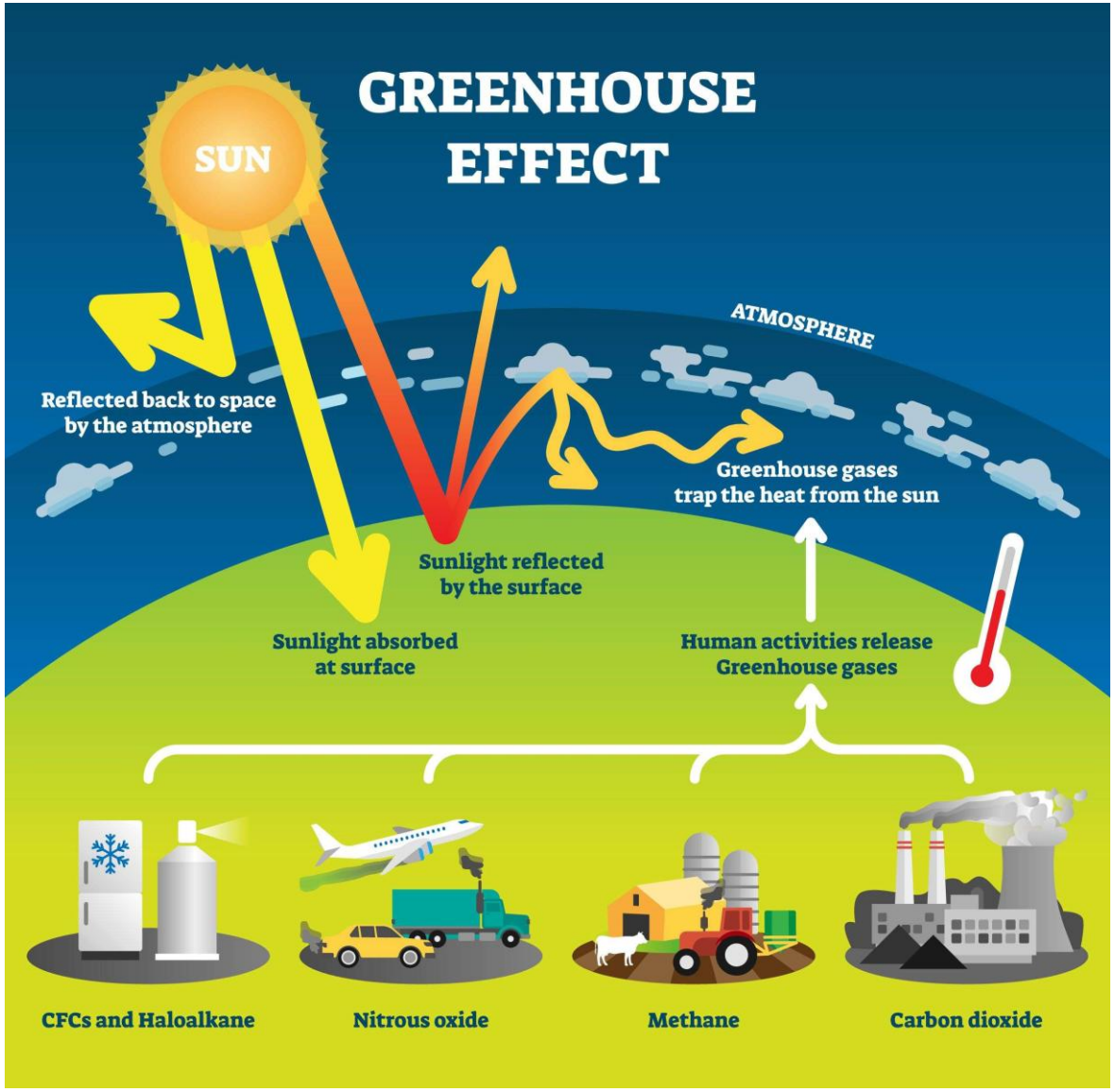
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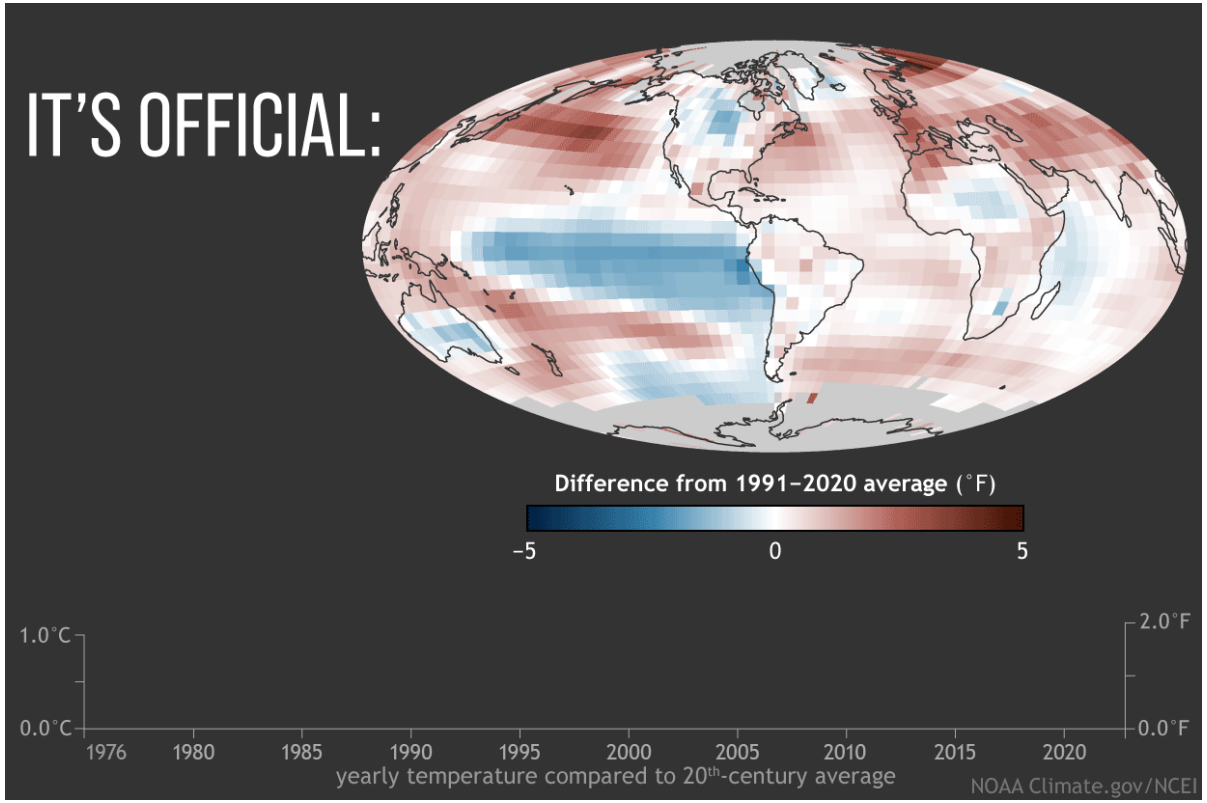
Ozone Layer 101



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Greenhouse Effect & Climate Change





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Climate Change – Greenhouse Effect

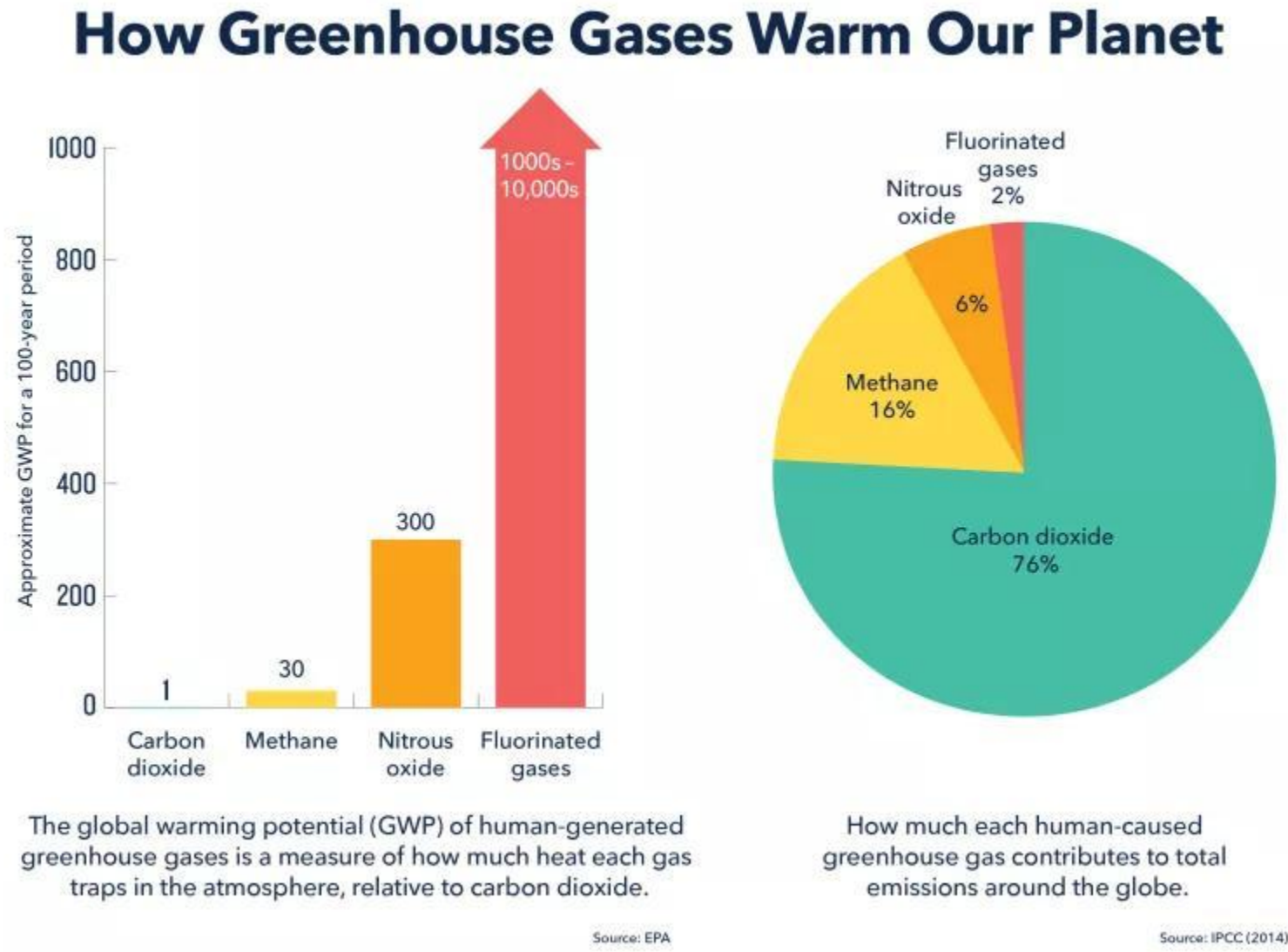
- Without greenhouse gases earth temperature **-6°C**
- With greenhouse gases earth temperature is **15°C**

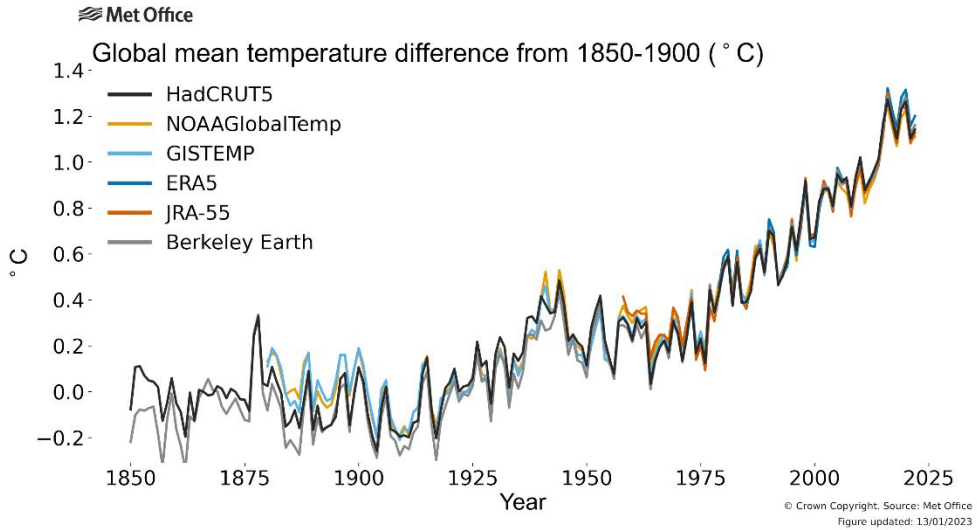
	Contribution to Enhanced Greenhouse Effect	Pre-industrial concentration	2015 concentration	Relative Strength as a greenhouse gas (per molecule)
Carbon dioxide	63%	280 ppmv	400 ppmv	1
Methane	18%	0.7 ppmv	1.8 ppmv	21
Nitrous oxide (N_2O)	6%	0.27 ppmv	0.32 ppmv	290
Others*	13%	0	<0.001 ppmv	> 3500

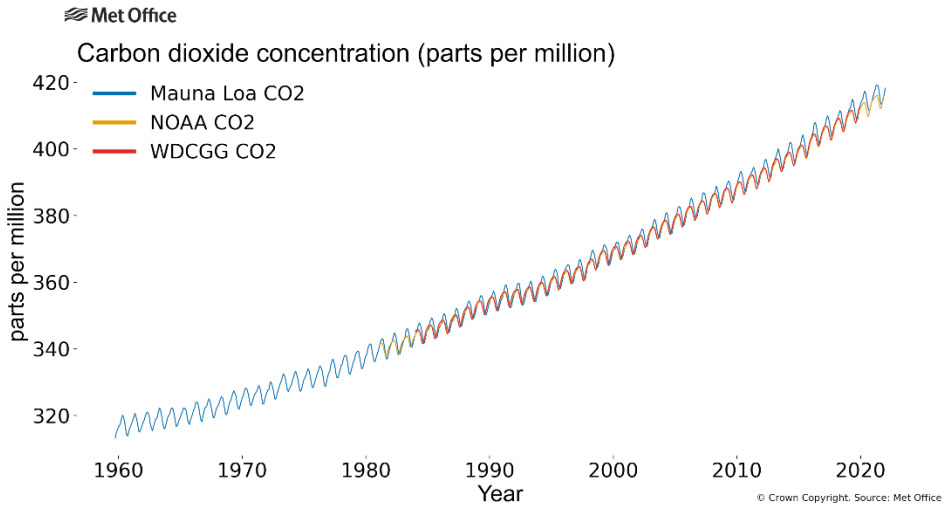
- Enhanced greenhouse effect is mostly caused by:
utilisation of fossil fuels for energy

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Greenhouse Gases







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Emissions from Energy Use

	kg carbon in 1kg of fuel	Energy (MJ) from 1 kg of fuel (GCV)	CO_2 from 1kg of fuel (kg)	CO_2 produced from 1 GJ of useful heat delivered* (kg)
Natural Gas	0.69	49.6	2.53	64
Oil (typical light fuel oil)	0.87	45.3	3.19	88
Coal (typical for power generation)	0.60	25.3	2.21	110
Electricity				65 in 2020 (was **230 in 1990**)
				* Assuming 80% efficiency (GCV)

In UK 95% of CO_2 & 80% of Greenhouse Gas emissions are from use of energy

Data from UK Gov statistics 2020

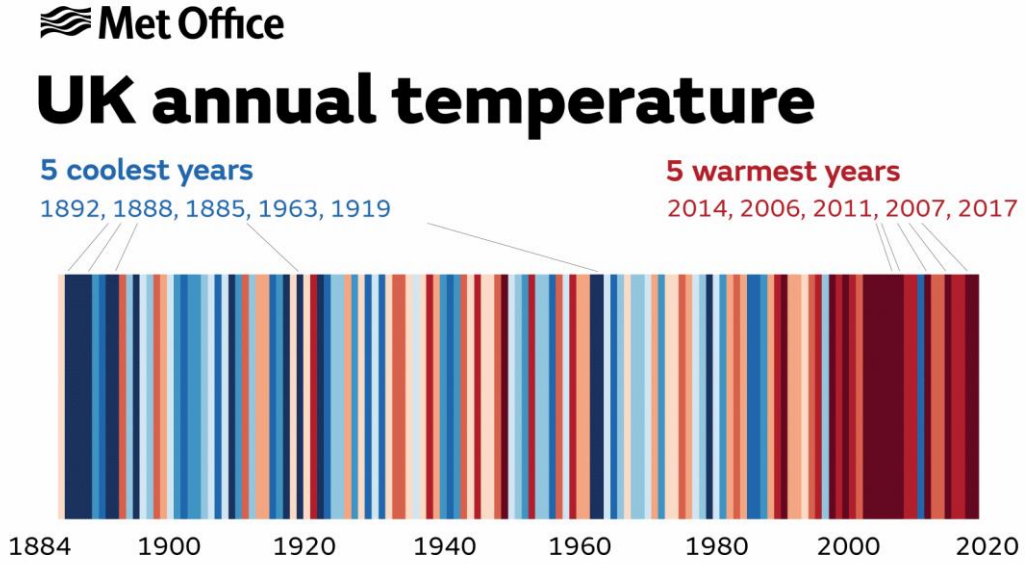
Impacts of Climate Change

Changes to the Climate System:

- **Rising ocean levels** – Rising temperatures are causing glaciers and ice sheets to melt, adding more water to the oceans and causing the ocean level to rise. Oceans absorb 90% of the extra heat from global warming: warmer water expands, and so our oceans are taking up more space.

- **Ocean acidification** – Ocean acidification occurs when the ocean absorbs carbon dioxide and becomes more acidic. It is often called the 'evil twin' of climate change.

- **Extreme weather events** – Climate change is causing many extreme weather events to become more intense and frequent, such as heatwaves, droughts, and floods.



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Impacts of Climate Change

Climate change can also affect people and ecosystems:

- **Flooding of coastal regions –** Coastal cities are at risk from flooding as sea levels continue to rise.

- **Food insecurity** – High temperatures, extreme weather events, flooding, and droughts can damage farmland. This makes it difficult for farmers to grow crops and means that their yield of crops each year is uncertain.

- **Conflict and climate migrants –** Climate change is a stress multiplier – it can take existing problems, such as lack of food or shelter, and make them worse. This can cause people to fight over resources (food, water, and shelter), or to migrate.

- **Damage to marine ecosystems –** Rising ocean temperatures, ocean acidification, and ocean anoxia (lack of oxygen) are damaging to marine life such as fish and coral reefs.

The Paris Agreement

In 2015, almost every country in the world signed a document promising to cut down on greenhouse gas emissions.

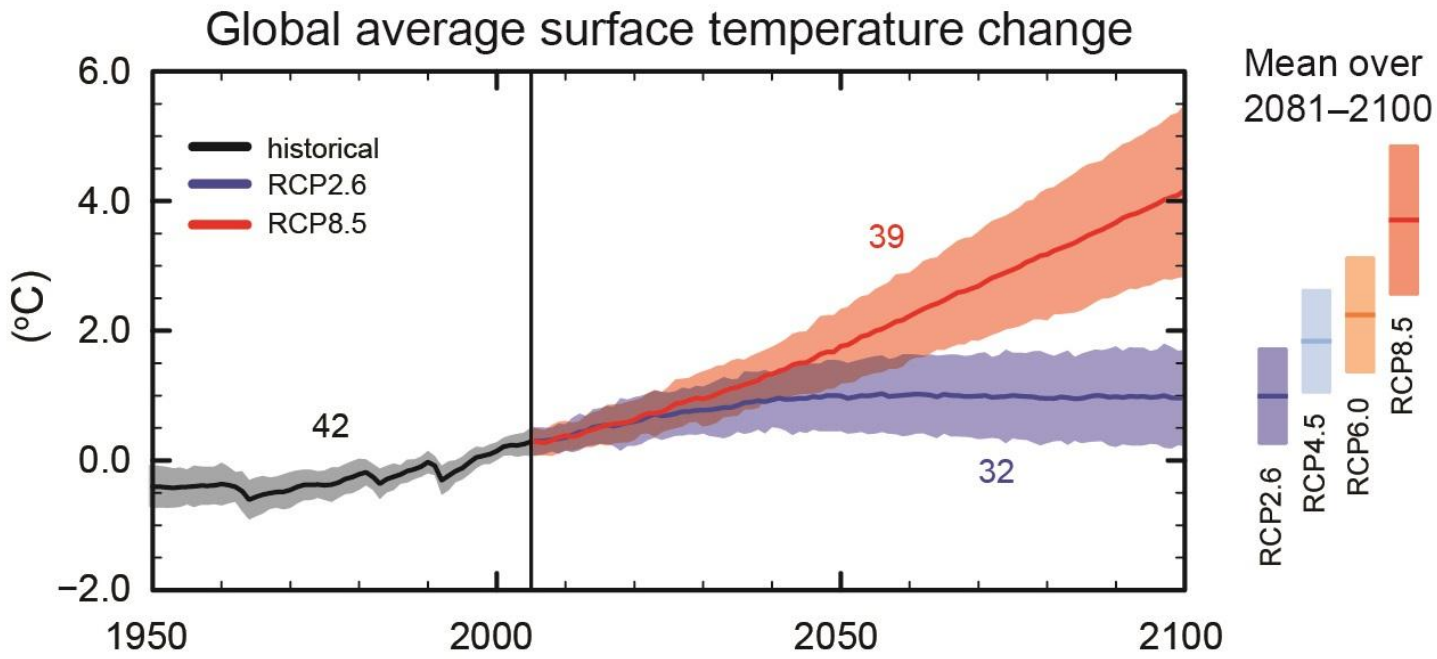
The aim was to limit the average global temperature to 2°C above pre-industrial temperatures.

If possible, countries pledged to aim for a 1.5°C limit.

Since then, the IPCC has published a report explaining the different impacts between a 1.5 or 2°C temperature rise.

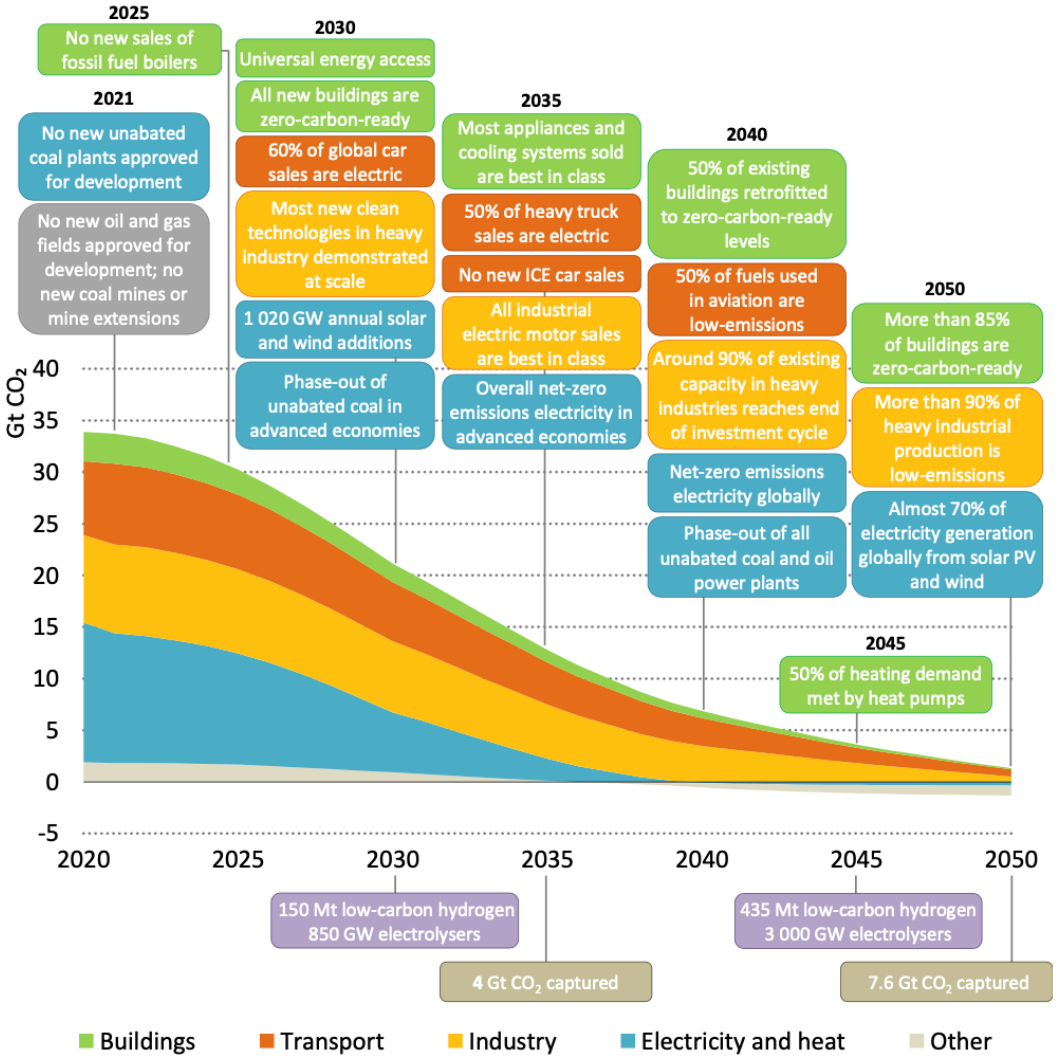
It showed that there are many benefits for people all over the world in limiting temperatures to 1.5°C.

Large and rapid reductions in global greenhouse gas emissions are needed to meet this goal.



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UK Net Zero Strategy



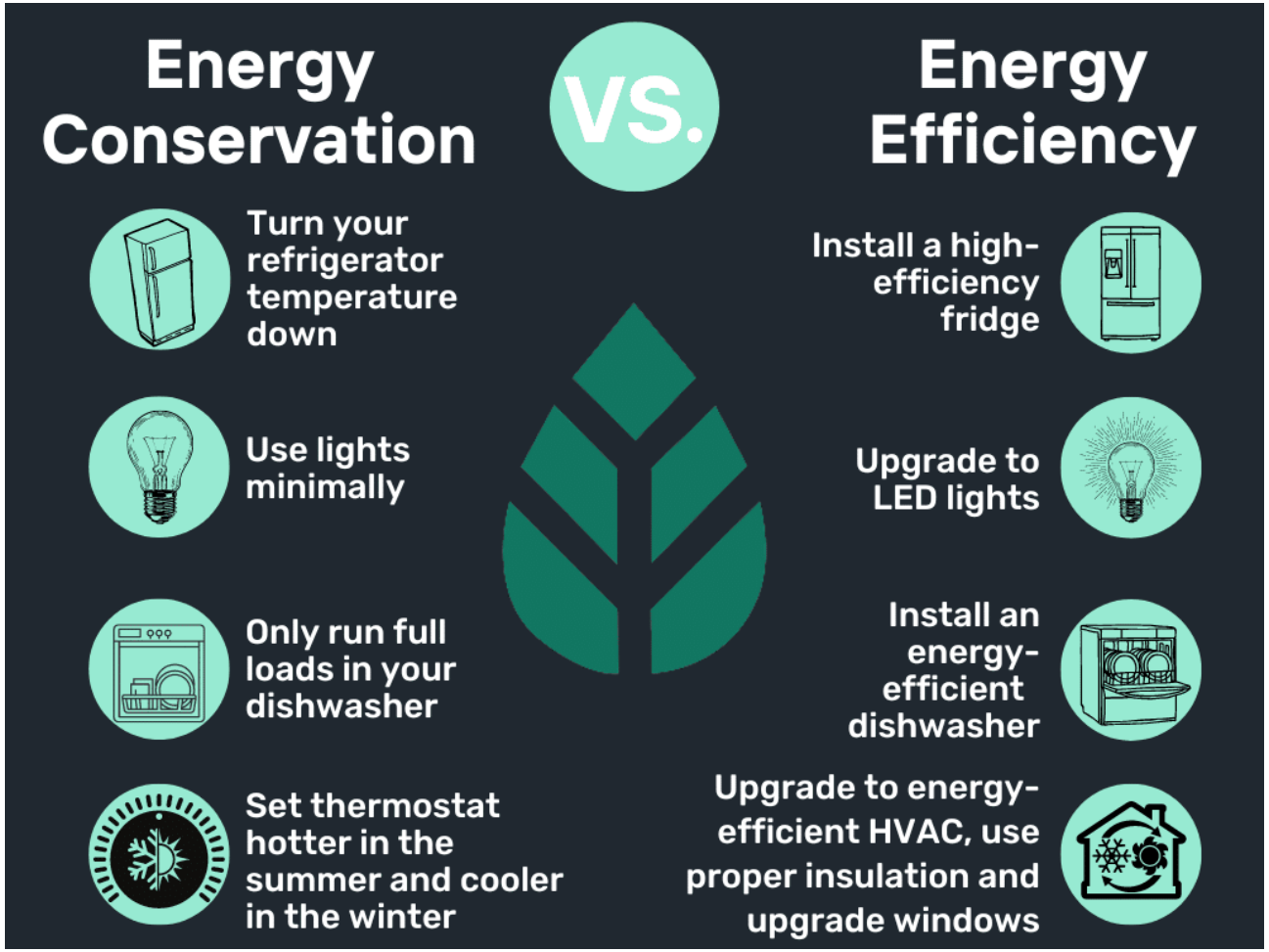
https://www.iea.org/reports/net-zero-by-2050

- By 2050, the UK is legally required to have reduced its greenhouse gas emissions by 100% from 1990 levels. The net zero target was made legally binding by the Climate Change Act 2008 (2050 Target Amendment) Order 2019.
- If met, this would mean the amount of greenhouse gas emissions produced by the UK would be equal to or less than the emissions removed by the UK from the environment.
- To set out how it planned to achieve this target, the government published the '[Net zero strategy: Build back greener]' in October 2019. This contained policies and proposals for decarbonising all sectors of the UK economy and meeting the net zero target by 2050.

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Energy efficiency measures

- Efficient Supply of Energy:
  - Efficient combustion of fuels
  - Use of combined heat and power
  - More efficient generation of electricity
  - Use of heat pumps to supply heat
- Reduce Demand for Energy:
  - Better insulation of buildings/industrial processes
  - More energy efficient industrial processes
  - More efficient transport – more public transport, reduce vehicle weight..
  - More efficient domestic appliances
  - Heat recovery systems



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Reducing greenhouse gas emissions (GHG) is not necessarily the same as saving energy!

- Reducing energy use will generally reduce greenhouse gas emissions.
- But switching to a lower carbon energy source will save greenhouse gases without necessarily reducing energy use (e.g. biofuels).
- Carbon emissions from electricity generation are reducing fast.

e.g. Two scenarios for investment in energy (likely to cost about the same):
- Improve insulation in your house and buy a higher efficiency gas fired boiler, or
- Buy a biofuel boiler - may be less efficient but could reduce GHG emissions more?

Suggest two scenarios for tropical climates?



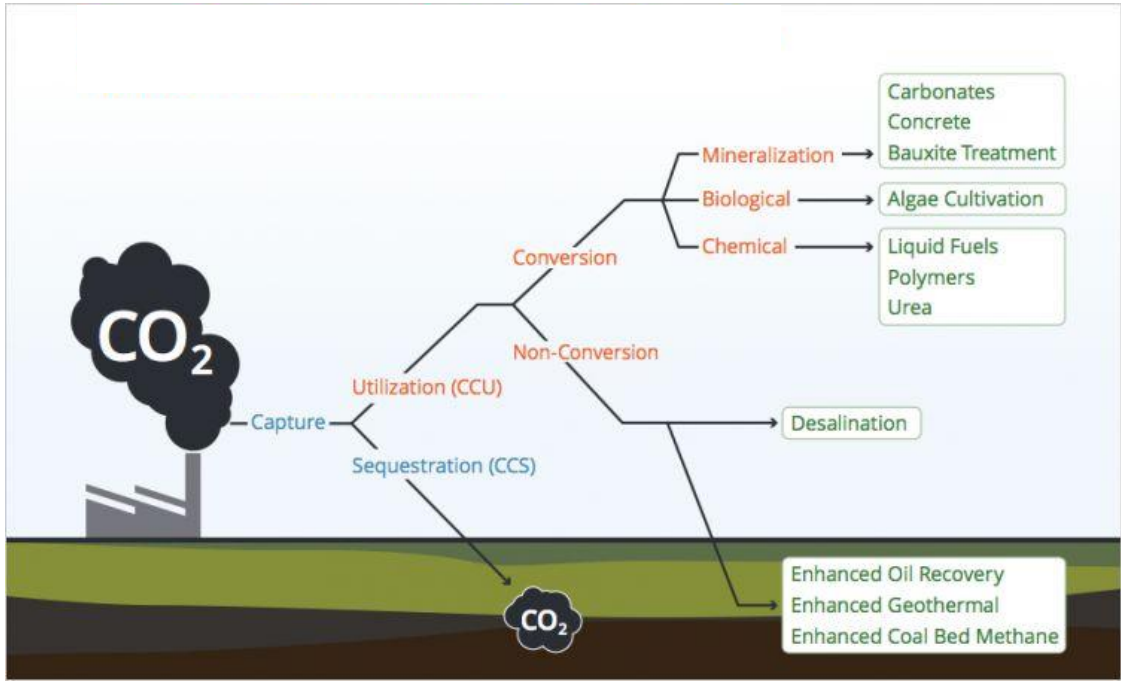
Removing Carbon: Carbon Capture & Storage

What is carbon capture, usage and storage (CCUS)?

CCUS refers to a suite of technologies that enable the mitigation of carbon dioxide (CO_2) emissions from large point sources such as power plants, refineries and other industrial facilities, or the removal of existing CO_2 from the atmosphere.

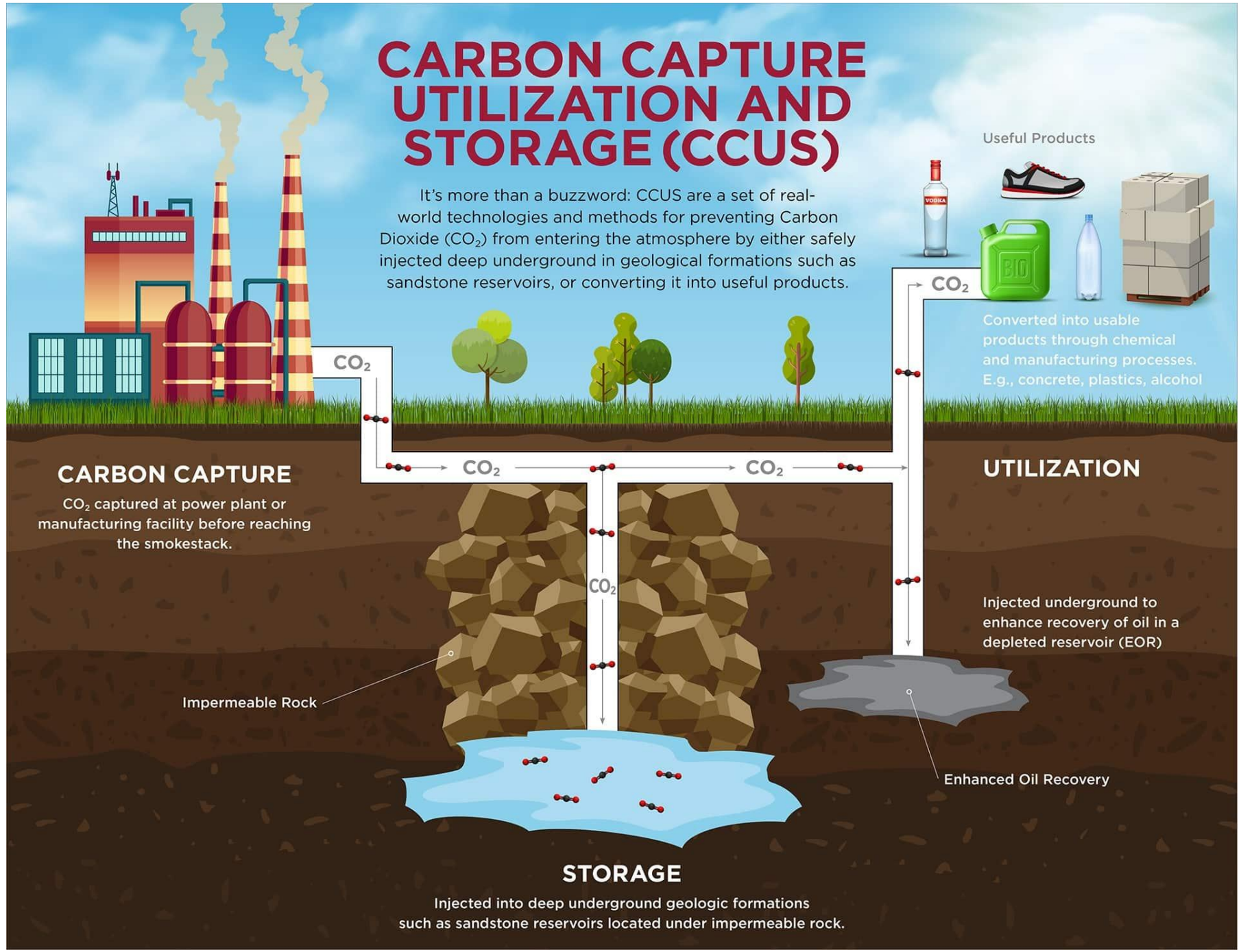
CCUS is expected to play a crucial role in meeting global climate targets.

Leading organisations including the IEA, IRENA and IPCC have all produced long-term energy outlooks that rely on a rapid expansion of CCUS in order to limit global temperature rise to 1.5°C.



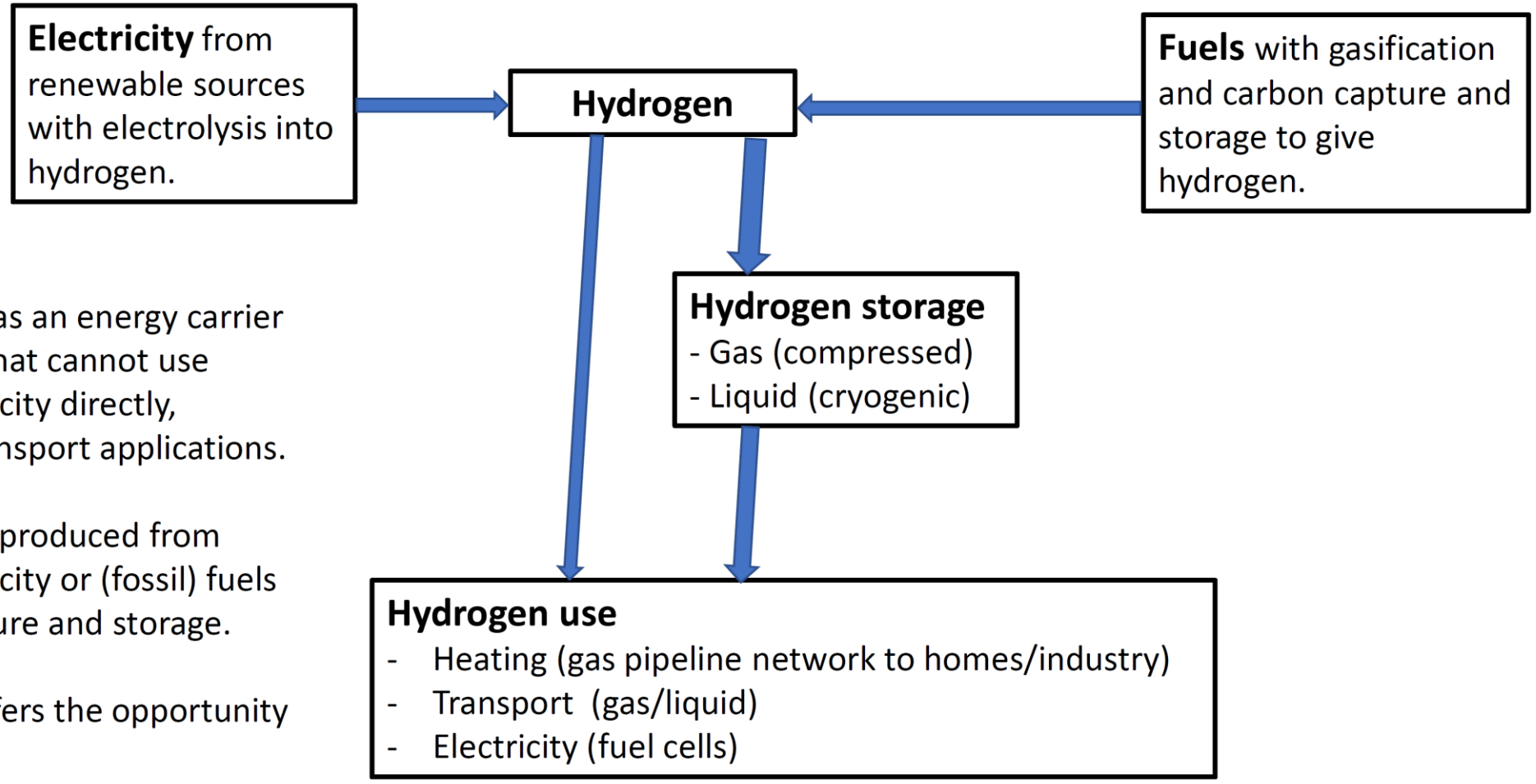
https://www.pembina.org/blog/technologies-behind-carbon-utilization

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Hydrogen for Energy Transport and Storage



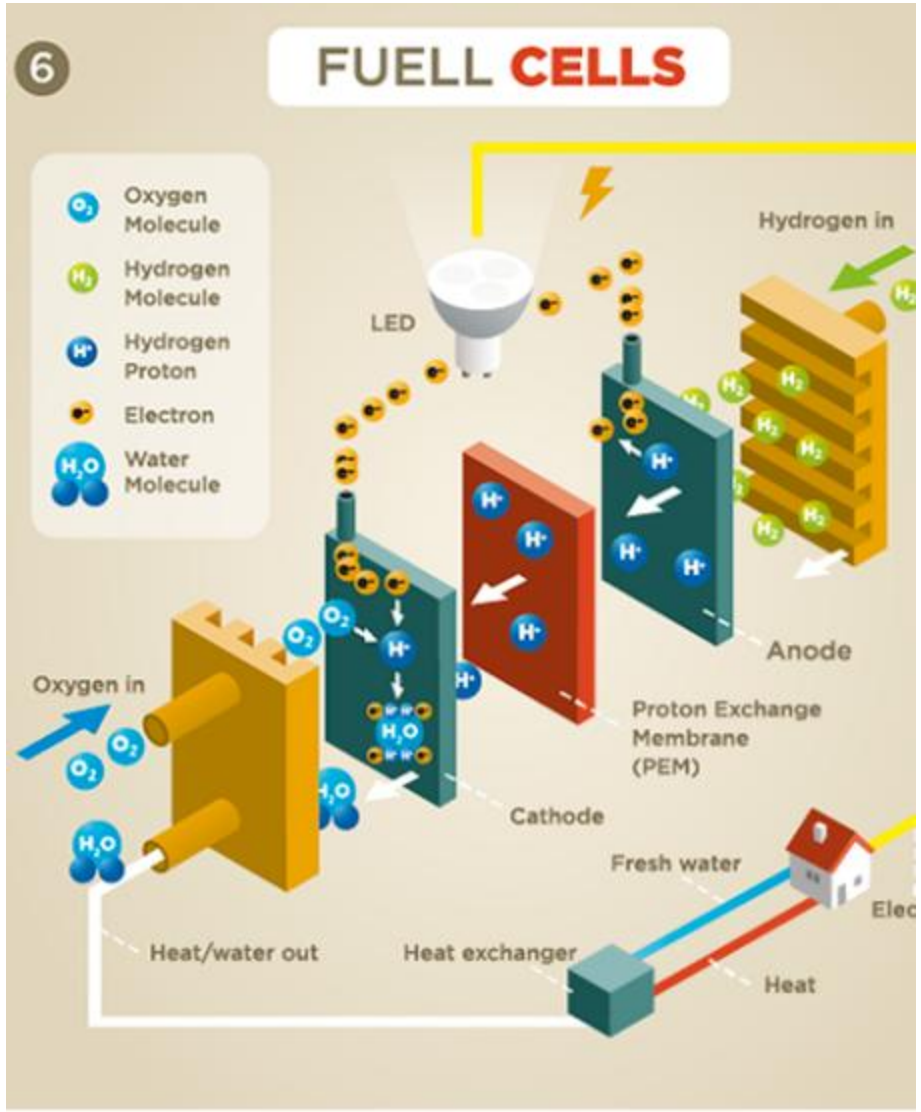
Use of hydrogen as an energy carrier for applications that cannot use renewable electricity directly, particularly in transport applications.

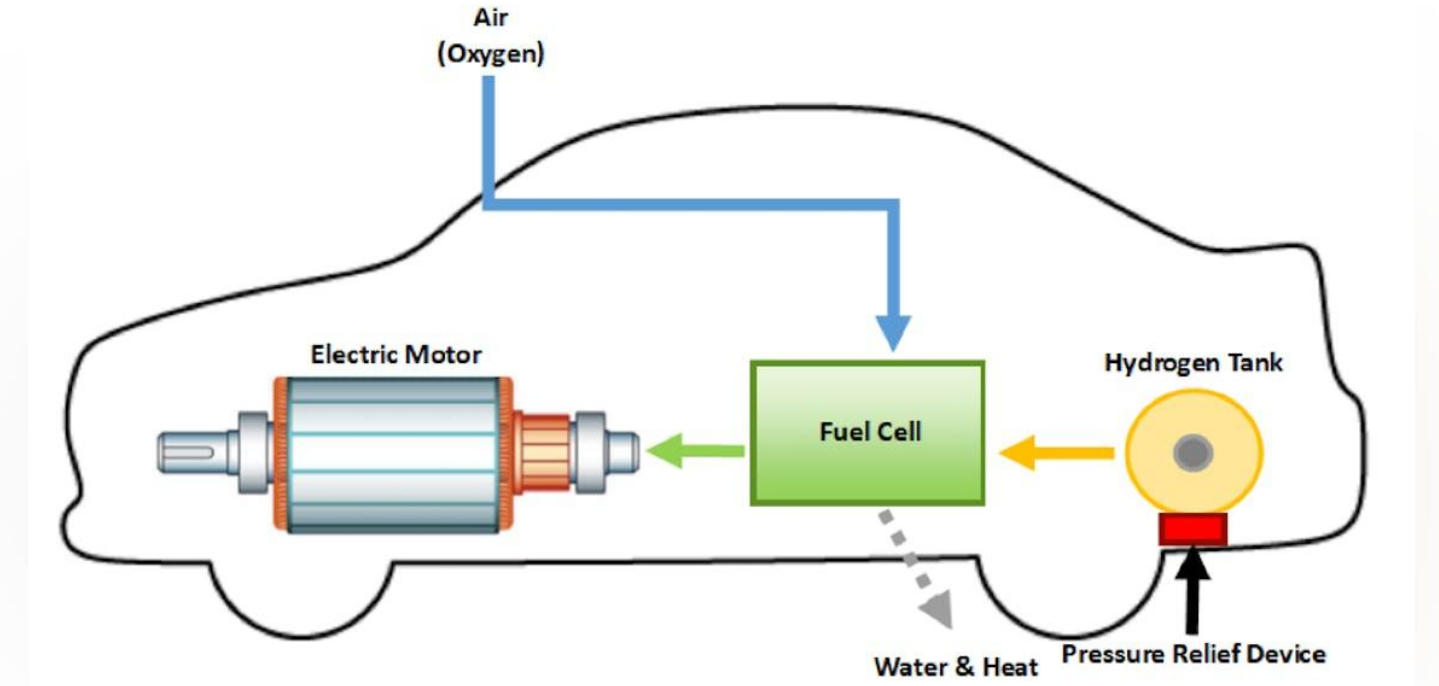
Hydrogen can be produced from renewable electricity or (fossil) fuels with carbon capture and storage.

Hydrogen also offers the opportunity to store energy.

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Fuel Cell





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Summery of learning outcomes

- ❑ Understand the origin, types, and global consumption trends of fossil fuels and their role in meeting global energy demand.
- ❑ Explain the fundamental principles of combustion in solid, liquid, and gaseous fuels, including oxidation reactions and heat release.
- ❑ Identify key atmospheric pollutants and greenhouse gases generated from energy use and combustion processes.
- ❑ Analyse the link between energy consumption, carbon emissions, and climate change, including historical and current emission patterns.
- ❑ Recognise the environmental and societal impacts of fossil fuel use, such as air pollution, acid rain, ozone depletion, and global warming.
- ❑ Evaluate strategies to reduce emissions and environmental impacts, including energy efficiency, cleaner combustion, carbon capture and storage, and alternative energy carriers such as hydrogen.
- ❑ Appreciate the role of international and national climate policies in driving the transition towards low-carbon and sustainable energy systems.